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Iron Ore Global New Capacity Projects in 2026
海外新增铁矿产能项目详解

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摘要 Abstract

Compared with the market expectations at the start of 2025, the commissioning pace of iron ore capacity has been delayed, but the capacity will most likely still be released in a phased manner. This paper sorts out 16 overseas mine projects that are expected to bring about supply increments in 2026, helping investors gain a clear understanding of the overall iron ore supply trend in 2026.

If the capacity is released as scheduled, the supply increment of overseas mines will reach 85 Mt; even under a relatively neutral scenario, the increment will still hit 47.5 Mt. In addition, these new capacity projects feature low production costs and thus enjoy strong competitive advantages.

Risks: This paper compiles, to the best of its ability, the new capacity projects that are likely to drive supply increments in 2026. The impact of mine replacement has been factored in for some companies, though the completeness of the statistics cannot be guaranteed. The commissioning schedule is subjectively inferred based on the information disclosed by the companies. The cash/operating costs of some projects are rough estimates and are for reference only.

较 2025 年初市场预期，铁矿产能投放节奏延后，但大概率仍将陆续释放。本文梳理了海外矿山预计将在 2026 年释放供应增量的 16 个项目，帮助投资者明晰铁矿石 2026 年的整体供应趋势。

若产能正常释放，海外矿山供给增量达 8500 万，相对中性预期下仍将达 4750 万吨，且新增产能项目成本较低，具有较强竞争优势。

风险提示：本文尽可能梳理了 2026 年可能带来供应增量的新增产能项目，部分公司考虑了矿山替代的影响，但不保证统计完全；投产进度在公司披露信息的基础上进行了主观推断；部分项目现金/运营成本为粗略估算，仅供参考。

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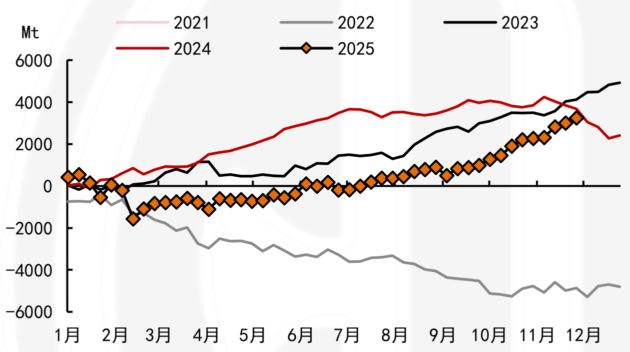
1. Overview

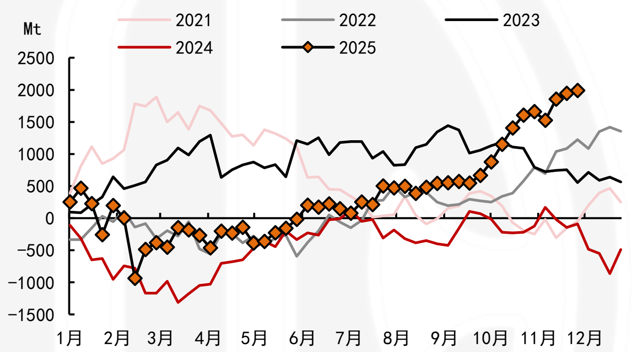
The global iron ore supply increment fell short of expectations in 2025. Affected by weather disruptions and slower-than-expected commissioning in the first half of the year, shipments registered a y/y decline before gradually recovering in the second half. As of early December, the total global shipments had increased by around 32 Mt year on year. The shipment growth was mainly driven by non-mainstream regions in Australia and Brazil, with the Onslow project seeing a marked production ramp-up in particular. Among the Four Major Mines, only FMG Iron Bridge project contributed a substantial supply increment, while shipments from non-mainstream countries dropped year on year.

图表1：Y/Y Difference in Cumulative Shipments of Global Mines



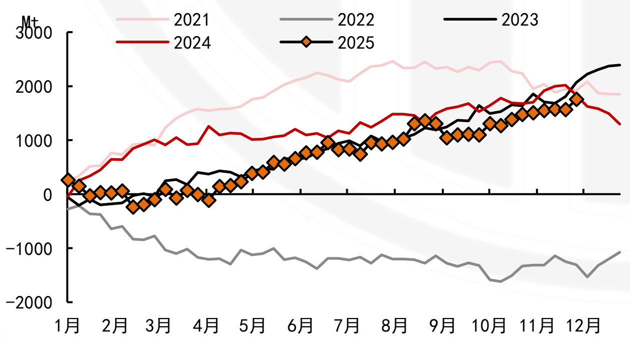
Source: Company's website, Mysteel, CITIC Futures

图表2：Y/Y Difference in Cumulative Shipments of Australian Mines



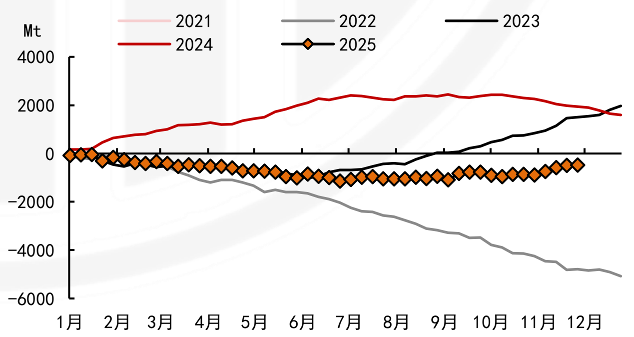
Source: Company's website, Mysteel, CITIC Futures

图表3：Y/Y Difference in Cumulative Shipments of Brazilian Mines



Source: Company's website, Mysteel, CITIC Futures

图表4：Y/Y Difference in Cumulative Shipments of Non-mainstream Countries Mines



Source: Company's website, Mysteel, CITIC Futures

Compared with the market expectations at the start of 2025, the commissioning pace of iron ore capacity has been notably delayed, yet the capacity will most likely still be released in a phased manner. This paper sorts out 16 overseas mine projects that are expected to bring about supply increments in 2026, helping investors gain a clear understanding of the overall iron ore supply trend for that year.

There is a substantial amount of potential new overseas capacity, and the global iron ore supply balance is turning looser. According to incomplete statistics, a considerable volume of new iron ore capacity will still be commissioned overseas in 2026. If these capacities come on stream as scheduled, overseas iron ore mines will see a y/y increase of 85 Mt in 2026. However, there may be a high incidence of unexpected disruptions during the initial phase of capacity commissioning. Back at the start of 2025, the market widely anticipated a large-scale capacity rollout, yet the actual supply increment fell short of expectations. Factoring in the risks of delayed capacity commissioning and the effect of mine capacity replacement, the y/y supply increment from overseas mines will still reach 47.5 Mt under a relatively neutral scenario, resulting in a relatively loose supply environment.

The cost of the new capacity remains at a low level, maintaining a distinct advantage over domestic mines and high-cost non-mainstream mines. Based on project disclosures and rough estimates, the cash costs of the new capacity projects compiled in this paper are mostly in the range of \$20 – \$50 per tonne, all below \$60 per tonne. These projects fall within the bottom 50% of the global cost curve, endowing them with a prominent cost advantage.

图表5：Overseas Iron Ore Supply Increment

	Project’ s newly added capacity	Capacity launch progress (plan)	Optimistic Scenario: Supply Increment	Neutral Scenario: Supply Increment	C1 Cash Cost
Australia		-	**2750**	**1650**	
FMG: Iron Bridge Magnetite Project	Newly added 22 Mt	First production achieved in 2023, with capacity ramping up	500	300	33-38美元/湿吨
Rio: Western Range Iron Ore Project	Newly added 25 Mt	Already commissioned, with production volume ramping up.	500	250	23-24.5美元/湿吨
Fenix: Weld Range Project	Newly added 8.6 Mt; phased capacity expansion from 1.4 Mt to 10 Mt	Initial shipment of the sub-mine commenced in August 2025; other mines are undergoing capacity expansion.	250	150	45-53美元/湿吨
Hancock: McPhee Creek Project	Newly added 9.5-9.7 Mt	Target: 1st ore production in FY25 - 26; No commissioning info available yet.	500	250	现金成本预计较低
MinRes: Onslow Iron Project	Newly added 35 Mt	Annualized capacity: Fully operational	1000	700	29-37美元/吨
MinRes: Lamb Creek Project	Newly added 7.5 Mt	Start production: Early 2026 | Role: Capacity replacement transitional project			49-54美元/湿吨
Brazilian			**1250**	**750**	
Vale: Serra Sul+20 Project	Newly added 20 Mt; phased capacity expansion from 100 Mt to 120 Mt	Commissioning: H2 2026 | Purpose: Replace old mine	400	200	20.5-22美元/吨
Vale: Capanema Project	Newly added 15 Mt	In operation, with capacity still ramping up.	600	400	20.5-22美元/吨
Samarco Project	Newly added 7-8 Mt; phased capacity expansion from 7-8 Mt to 14-16 Mt	Phased capacity recovery target achieved; production ramping up; further recovery planned.	150	100	粗略测算可能在30-33美元/吨
CSN: Pires Tailings Recovery Project	Newly added 1.1 Mt	Commissioning: Q1 2026	100	50	预计低于20美元/吨
Guinea			**2000**	**1000**	
Simandou Project	Newly added 120 Mt	First shipment delivered; currently in the ramp-up phase.	2000	1000	预计高于24.5美元/吨
Liberia			**900**	**500**	
ArcelorMittal: Phase II Expansion Project of Nimba Iron Ore Mine	Newly added 15 Mt; phased capacity expansion from 5 Mt to 20 Mt	Production started in June 2025; capacity is ramping up.	900	500	約30美元/吨
South Africa			**150**	**100**	
Palabora: Phase II Deep Development Project	Newly added 3 Mt,Capacity Expansion Project	Commissioning: Jul 2026	150	100	低于全球主流铁矿项目
India			**1450**	**750**	
KIOCL: Devadari Project	Newly added 1.5 Mt; phased capacity expansion from 0.5 Mt to 2 Mt	Production volume surpass 1 Mt in FY2026 - 2027	50	50	
LMEL: Surjagarh Project	Newly added 16 Mt; phased capacity expansion from 10 Mt to 26 Mt	ramping up	1000	500	估算在40-50美元/吨附近
A series of auction-based resumption projects in Goa	No more than 20 Mt	A few projects operational	400	200	
Total	-	-	**8500**	**4750**	

Source: Company's website, Mysteel, CITIC Futures

2. Australian

2.1 FMG: Iron Bridge Magnetite Project

The Iron Bridge Magnetite Project is a high-grade magnetite ore project developed by FMG (Fortescue Metals Group), one of Australia’s leading mining companies. Located in the Pilbara region of Western Australia, it is among the world’ s few large-scale high-grade magnetite development projects. FMG Magnetite Pty Ltd holds a 69% stake in the project, while Formosa Steel IB Pty Ltd (an affiliate of Formosa Steel) owns the remaining 31%.Notably, FMG Magnetite Pty Ltd is a wholly-owned subsidiary of FMG Iron Bridge Ltd, in which FMG holds an 88% stake and Baoshan Iron & Steel Resources International Co., Ltd. owns a 12% stake.

The total investment of the Iron Bridge Magnetite Project is approximately 3.9 billion US dollars, with proven reserves of 716 Mt (at an ore grade of 67%) and total mineral resources of 5.448 billion tonnes (at an ore grade of 30.4%). It is expected to produce 22 Mt of high-grade magnetite concentrate with an iron content of 67% per annum.

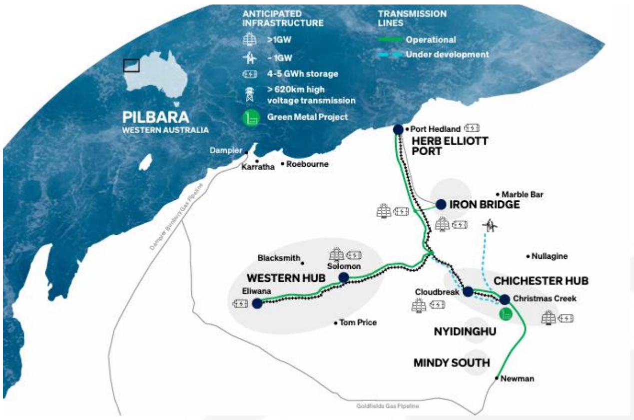
The products of the Iron Bridge Magnetite Project enjoy convenient transportation access. Located in the Pilbara region of Western Australia, the project site is 145 kilometers south of Port Hedland. It adopts a pipeline transportation system, consisting of a 135-kilometer slurry and return water pipeline as well as a 179-kilometer high-pressure glass-reinforced plastic (GRP) raw water pipeline. Wet concentrate is transported from the mine to Port Hedland via these pipelines, where it undergoes dehydration and material processing to be converted into high-grade magnetite products.

The cash cost of the Iron Bridge Magnetite Project is approximately $33－$38 per wet metric ton. In terms of operational costs, as previously reported by Steel Union, the project’s cash cost will remain in the range of $33－$38 per wet metric ton throughout its mining life, which includes fees for Port and Power Services provided by FMG. This cost advantage is also quite prominent when compared with domestic iron ore mines in China.

The capacity ramp-up of the Iron Bridge Magnetite Project is well underway. The project transitioned to operational production in August 2023 and shipped its first batch of high-grade (67% iron content) magnetite products in September 2023. The project is ramping up its capacity in phases. According to the latest plan released in Fortescue Metals Group’s (FMG) May 2025 announcement, Fortescue expects the annual shipment volume of the Iron Bridge Magnetite Project to reach 10-12 Mt in fiscal year 2026 (covering the calendar period from July 2025 to June 2026); this figure will rise to 16-20 Mt by fiscal year 2027 (July 2026 to June 2027). The ultimate goal is to achieve the full-capacity output of 22 Mt in fiscal year 2028 (July 2027 to June 2028), representing a nearly three-year delay from the previously targeted full commissioning date of September 2025. In the third quarter of 2025, the processing volume of the Iron Bridge Magnetite Project stood at 2.3 Mt, equivalent to an annualized processing capacity of 9.2 Mt.

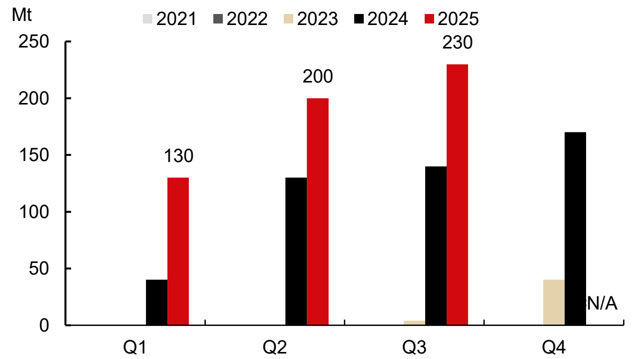
Supported by the Iron Bridge Magnetite Project, FMG is expected to see a supply increase of approximately 3-5 Mt in 2026. FMG’s total shipment guidance for fiscal year 2026 (FY26) stands at 195-205 Mt, representing a 5 million-tonne rise compared with fiscal year 2025 (FY25). This includes 10-12 Mt from the Iron Bridge (on a 100% basis), with the majority of the incremental shipment volume under the guidance attributable to the Tieqiao Project. The processing volume of the Iron Bridge reached 6.4 Mt in FY25. Assuming the quarterly annualized output grows linearly to the upper limit of the guidance target, the y/y shipment increment in 2026 is projected to hit 5 Mt; if it grows linearly to the lower limit of the guidance target, the y/y shipment increment is expected to reach 3 Mt.

图表6：Location of the Iron Bridge Magnetite Project



Source: Company's website, Mysteel, CITIC Futures

图表7：Quarterly Processing Volume of the Iron Bridge Magnetite Project



Source: Company's website, Mysteel, CITIC Futures

2.2 Rio：Western Range Iron Ore Project

The Western Range Iron Ore Project is one of the core projects under Rio Tinto's mine replacement program in the Pilbara region, designed to maintain the stable production capacity of its Pilbara Blend. Rio Tinto holds a 54% equity stake, while Baowu Steel Group owns the remaining 46%. Meanwhile, the West Slope Project is actually an extension of the Brockman Iron (East Ridge) Project. Jointly established by the former Baoshan Iron & Steel Group and Hamersley Iron Pty Ltd., a subsidiary of Rio Tinto, the East Range Project was launched to develop mineral deposits on the eastern slope of the Hamersley Range, with a maximum production capacity of 200 Mt—a threshold that was reached in October 2020.

The total investment of the Western Range Project amounts to 2 billion US dollars, which is jointly developed by Rio Tinto Group and China Baowu Steel Group. With a production capacity of 25 Mt per annum, the project is expected to yield a total of 275 Mt of iron ore at an average grade of approximately 62%, boasting a mine life of 20 years. Calculated on the basis of the 25-million-tonne annual output, the mine life would be around 11 years, but Rio Tinto has stated that the operational duration could be extended in accordance with market demand. In addition, Baowu and Rio Tinto have entered into an iron ore procurement agreement, under which Baowu will purchase a total of 126 Mt of iron ore products from Rio Tinto in proportion to its equity stake, equivalent to an annual procurement volume of approximately 11.5 Mt.

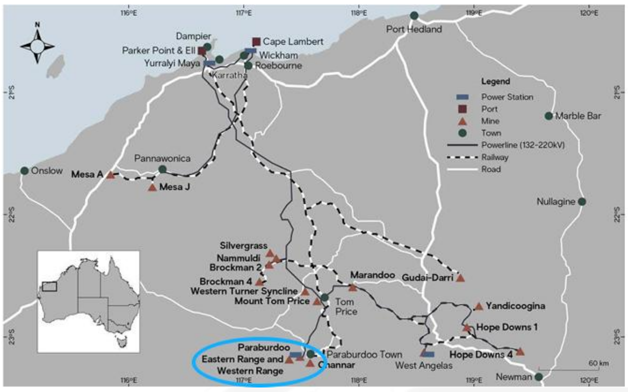
The Western Range Project is located in the Pilbara region of Western Australia, Australia. The new development involves the construction of a primary crusher and an 18-kilometer conveyor belt connected to the Paraburdoo Processing Plant, which will transport the crushed ore from the mine to the existing Paraburdoo Concentrator. Paraburdoo is one of Rio Tinto’s earliest mining hubs and has been in operation since 1972.

The unit cash cost of the Western Range Project is expected to range between $23 and $24.5 per wet metric ton. Leveraging Rio Tinto’s existing infrastructure in the Pilbara region, the project boasts a highly competitive unit cost. According to the overseas mining investment brief, the unit cash cost of West Slope iron ore is projected to fall within the range of $23-$24.5 per wet metric ton, which is consistent with Rio Tinto’s 2025 cost guidance range for the Pilbara region.

The Western Range Project has been commissioned and is in the phase of production ramp-up. It achieved its first ore production in March 2025, and is expected to reach its full design capacity by the end of 2025, at which point the annual production capacity will hit 25 Mt. Production volume is set to continue growing as scheduled for the remainder of 2025.

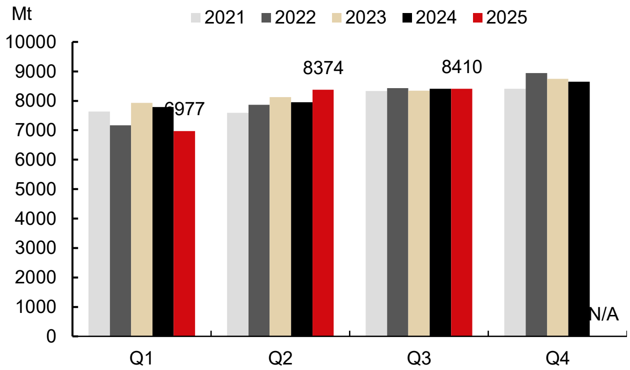
Rio Tinto’s supply increment is estimated to be around 2.5-5 Mt in 2026. The primary objective of the Western Range Project is mine replacement. Despite having been commissioned, it has not unlocked any notable supply growth, and will roughly offset the capacity gap arising from mine depletion. Rio Tinto disclosed that 11 Mt of production capacity in the Pilbara mining area will be exhausted in 2026. The production target for Rio Tinto’s Pilbara mining area in 2026 remains unchanged at 323-338 Mt, consistent with the 2025 target. It is expected that the 2025 output will meet the lower bound of the target range, while the 2026 shipment volume is projected to increase by 5 Mt y/y, with a neutral forecast of a 2.5 million metric ton y/y rise.

图表8：Location of the Western Range Project



Source: Company's website, Mysteel, CITIC Futures

图表9：Rio Tinto recorded no notable output growth in the third quarter



Source: Company's website, Mysteel, CITIC Futures

2.3 Fenix Resources: Weld Range Project

The Weld Range Project is a large-scale hematite iron ore project in Western Australia, jointly developed by Sinosteel Midwest Corporation Limited(SMC, subsidiary of China Baowu Steel Group) and Fenix Resources. Fenix has secured 100% of the revenue generated from iron ore sales of the Weld Range Project by paying a cash consideration of USD 60 million, followed by subsequent production royalties and profit royalties. Pursuant to the Beebyn-W11 10Mt RTMA signed in October 2023, Fenix obtained the right to mine 10 Mt of iron ore from the Beebyn-W11 deposit without paying production royalties or profit sharing. The Weld Range RTMA signed in September 2025 covers the remaining resource portion of the Weld Range Project.

The Weld Range Project is dominated by hematite ore, with total hematite resources of approximately 290 Mt and an average iron grade of 56.8%, classifying it as a medium-to-high grade hematite ore. It features low impurity content and can be directly transported and sold without complex beneficiation processes, qualifying it as a Direct Shipping Ore (DSO). Among its deposits, the Beebyn-W11 deposit has an estimated resource volume of 20.5 Mt, with an iron grade reaching 61.3%. The project plans to integrate existing production capacity and phase in capacity expansion from 1.4 Mt to 10 Mt.

The Weld Range Iron Ore Project has access to shared existing transportation infrastructure. Located in the Mid-West region of Western Australia, Australia, it is approximately 500 kilometers away from the Port of Geraldton and adjacent to Fenix Resources' existing Iron Ridge Project, enabling the sharing of logistics infrastructure including the following: 1)Newhaul Road Logistics: The company owns and operates a state-of-the-art road freight fleet, and also has two dedicated rail sidings located in Ruvidini and Perenjori respectively. 2)Newhaul Port Logistics: The company owns and operates three bulk cargo storage sheds at the Port of Geraldton, with a total storage capacity of over 400,000 metric tons and an annual export loading capacity of more than 10 Mt.

The unit cash cost of the Weld Range Project is approximately $45－$53 per wet metric ton. Benefiting from low mining costs and access to shared existing transportation infrastructure, and with reference to the production costs of Fenix Resources’ existing operations, the C1 cash cost guidance on a FOB Port of Geraldton basis for Fiscal Year 2026 is AUD 70－80 per wet metric ton, equivalent to approximately $45－$53 per wet metric ton.

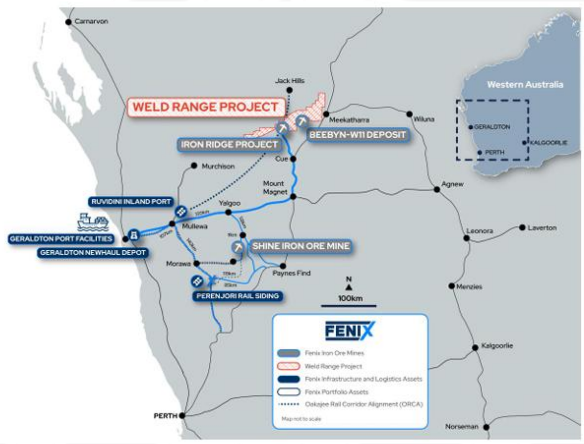
The production capacity of the Weld Range Project is being gradually realized. In September 2025, SMC and Fenix signed a 30-year mining contract agreement, marking the project’s entry into the substantive development phase. Fenix plans to complete the feasibility study report within 12 months of the Execution Date (September 2026) and commence commercial production within 24 months (September 2027).Currently, Fenix has launched the feasibility study for the Weld Range Project, which will integrate resources from two parts: first, the production volume currently generated by Fenix from the Iron Ridge Iron Ore Mine and the Beebyn-W11 Iron Ore Mine; second, the high-grade hematite Direct Shipping Ore (DSO) deposits that constitute the remaining resources of the Weld Range Project.

The initial priorities of the feasibility study include accelerating and expanding the existing mining plan for Beebyn-W11, as well as initiating new mining operations at the adjacent Beebyn-W10 and Madoonga-W14 deposits. The Madoonga mining area is set to become the primary hub supporting the annual production capacity of 10 Mt in the future. Fenix commenced mining operations at Iron Ridge (originally its wholly-owned iron ore project with a capacity of 1.4 Mt) in December 2020, and launched operations at Beebyn-W11 (a sub-project of Weld Range with a capacity of 1.5 Mt) in June 2025. The first batch of iron ore from Beebyn-W11 has already been shipped. In Q3 2025, Fenix achieved a production capacity of 3.5 Mt, and it aims to hit the annual target capacity of 4 Mt by the end of the year.Pursuant to the Weld Range RTMA, Fenix has a performance obligation to maintain an annual output of at least 6 Mt at Weld Range after the completion of the production ramp-up phase. Meanwhile, the company has also entered into an agreement with China Baowu Steel Group, under which the two parties will collaborate on the targeted export and sale of 10 Mt of iron ore per annum.

Fenix’s supply increment is expected to reach 1.5－2.5 Mt in 2026. The Weld Range Iron Ore Project of Fenix Resources has a planned annual capacity of 10 Mt. Its sub-project, the 1.5-million-metric-ton Beebyn-W11 Iron Ore Mine, completed its first shipment on August 18, 2025.

The Madoonga Mining Area, the project’ s flagship mining zone, is in the development preparation phase. Currently, the overall feasibility study of the project is underway, focusing on the integration of existing infrastructure and the development plan for new mining areas. The project is scheduled to complete its production ramp-up by 2028 and achieve the first export of 10 Mt of iron ore. It is estimated that the project will unlock an incremental supply of 2.5 Mt in 2026, with a neutral forecast of a 1.5-million-metric-ton y/y increase.

图表10：Location of the Weld Range Project



Source: Company's website, Mysteel, CITIC Futures

图表11：Resource Volume of The Weld Range Project

Deposit	Classification	Tonnes (Mt)	Fe (%)	SiO_2 (%)	Al_2O_3 (%)	LOI (%)	P (%)	S (%)
Madoonga	Measured	63.9	57.42	6.85	2.20	7.88	0.08	0.09
	Indicated	41.9	54.98	10.14	2.13	7.95	0.08	0.14
	Inferred	13.3	54.01	12.11	2.10	7.51	0.07	0.13
	Total (Mes+Ind)	105.9	56.45	8.16	2.17	7.91	0.08	0.11
	Total (Mes+Ind+Inf)	119.1	56.18	8.60	2.17	7.86	0.08	0.11
Beebyn	Measured	58.0	58.30	6.19	2.72	6.63	0.11	0.02
	Indicated	23.6	57.21	7.56	3.07	6.53	0.10	0.02
	Inferred	7.2	54.75	9.77	4.01	6.77	0.10	0.03
	Total (Mes+Ind)	81.6	57.99	6.59	2.82	6.60	0.11	0.02
	Total (Mes+Ind+Inf)	88.8	57.72	6.85	2.92	6.62	0.11	0.02
Beebyn North	Measured	7.3	55.74	8.91	1.95	8.67	0.08	0.06
	Indicated	16.6	54.19	11.79	2.08	7.65	0.09	0.07
	Inferred	28.6	55.56	11.57	2.15	5.86	0.09	0.19
	Total (Mes+Ind)	23.9	54.66	10.91	2.04	7.96	0.09	0.07
	Total (Mes+Ind+Inf)	52.5	55.15	11.27	2.10	6.82	0.09	0.14
Madoonga West	Measured	-	-	-	-	-	-	-
	Indicated	-	-	-	-	-	-	-
	Inferred	5.3	53.81	12.68	1.91	7.90	0.12	0.03
	Total (Mes+Ind)	-	-	-	-	-	-	-
	Total (Mes+Ind+Inf)	5.3	53.81	12.68	1.91	7.90	0.12	0.03
W38/39	Measured	-	-	-	-	-	-	-
	Indicated	-	-	-	-	-	-	-
	Inferred	3.2	54.90	13.60	1.76	5.75	0.07	0.05
	Total (Mes+Ind)	-	-	-	-	-	-	-
	Total (Mes+Ind+Inf)	3.2	54.90	13.60	1.76	5.75	0.07	0.05
	Total (Mes+Ind)	232.0	57.24	7.52	2.43	7.03	0.09	0.07
	Total (Mes+Ind+Inf)	290.3	56.77	8.35	2.42	6.93	0.09	0.08

Note: 50% Fe cut-off

Source: Company's website, Mysteel, CITIC Futures

2.4 Hancock Prospecting: McPhee Creek Project

The McPhee Creek Iron Ore Project is a small-to-medium-sized iron ore project in the Pilbara region, developed by Atlas Iron, a subsidiary of Hancock Prospecting of Australia. On July 1, 2025, Hancock Prospecting merged its 70%-owned Roy Hill Iron Ore and 100% - owned Atlas Iron to establish Hancock Iron Ore. Following the merger, the McPhee Creek Iron Ore Project, as a core asset of Atlas Iron, was incorporated into the unified management of Hancock Iron Ore.

The investment amount of the McPhee Creek Project is approximately USD 600 million. The project plan includes 5 open-pit mines (including some underwater pits). The ore resources are mainly magnetite and hematite, with an average grade of about 58%–60% and a mine life of around 15 years. The designed annual output is 9.5–9.7 Mt of iron ore (some sources mention 8–10 Mt). The run-of-mine ore does not require independent beneficiation and needs to be transported to the Roy Hill Processing Plant for processing.

The McPhee Creek Project has access to existing infrastructure. Located in the northeastern part of the Pilbara region, Western Australia, Australia, it is approximately 100 kilometers north of

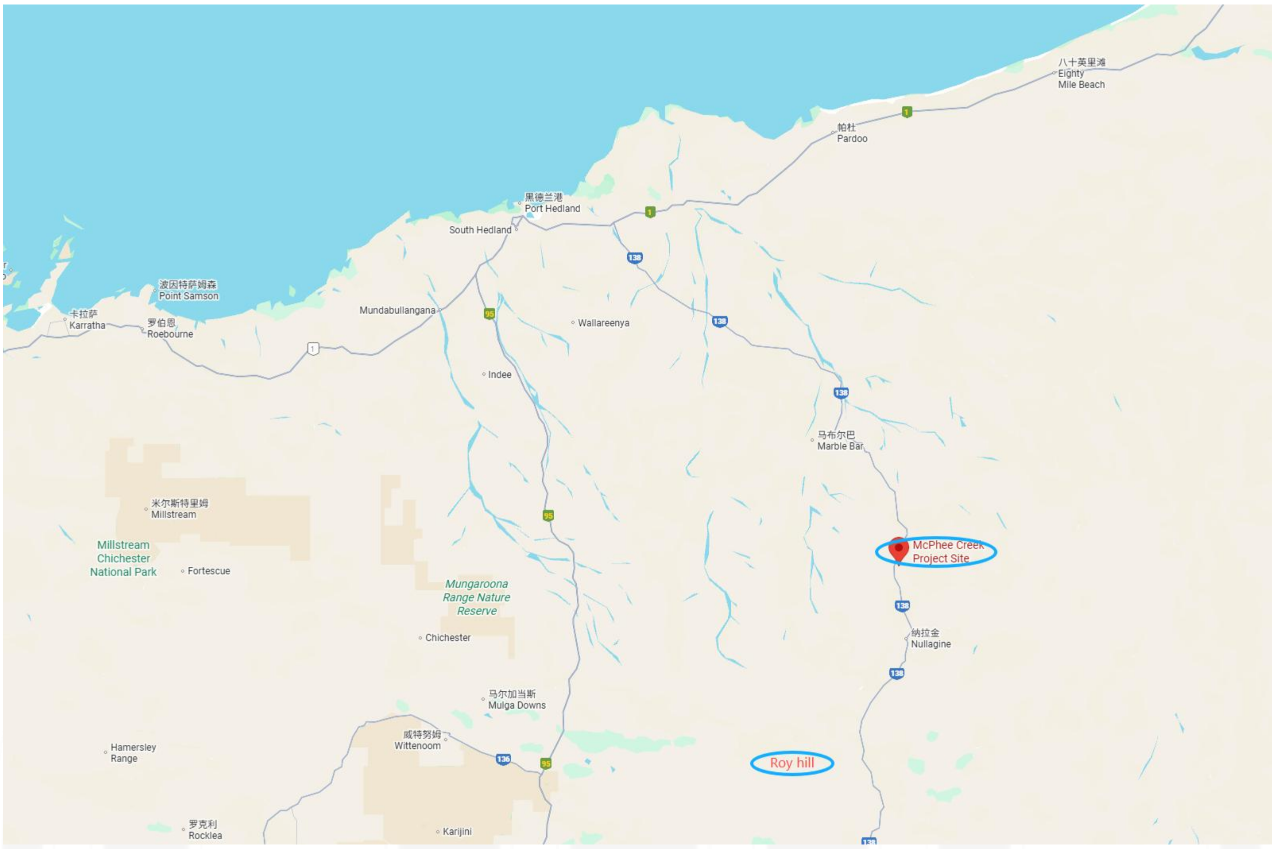
Hancock Prospecting's Roy Hill Iron Ore Project. Its core feature is achieving low-cost operations by leveraging Roy Hill's existing infrastructure. Adjacent to the existing highway network, the project can be easily connected to Roy Hill's logistics system.After ore extraction, the raw ore is initially crushed on-site at a 10-million-metric-ton-per-annum primary crushing plant operated by CSI Services, a subsidiary of Mineral Resources. It is then transported via road trains to the Roy Hill Iron Ore Processing Center 100 kilometers away. Subsequent processes such as blending and desliming are completed using Roy Hill's existing equipment. Finally, the processed ore is transported via Roy Hill's 344-kilometer dedicated railway to Port Hedland for export, eliminating the need to construct new processing facilities, railways, or port infrastructure.

The McPhee Creek Project is expected to boast a relatively low unit cash cost. By sharing the infrastructure of Roy Hill, the project eliminates the need for substantial capital expenditure on infrastructure construction, thus achieving a more competitive cash cost control. Drawing on the cost level of Roy Hill, the C1 cash cost of the McPhee Creek Project is projected to rank in the lower-mid range of the industry. No specific figures have been disclosed, but it is expected to be lower than that of the Roy Hill Project.

The McPhee Creek Project is scheduled to commence production in the near term, though no concrete implementation details have been released yet. Initially slated for commissioning in 2023, the project was delayed due to approval hold-ups. It was later planned to produce its first batch of ore in the 2025–26 fiscal year, marking its official entry into the operational phase. To date, no commissioning updates have been announced.

The McPhee Creek Iron Ore Project is expected to achieve a supply increment of 2.5–5 Mt in 2026. If commissioning proceeds smoothly in 2025, it is expected to unlock a supply increment of 5 Mt in 2026; if commissioning takes place in mid-2026 (the end of the 2026 fiscal year), the supply increment is projected to reach 2.5 Mt.

图表12：Location of the McPhee Creek Project



Source: Company's website, Mysteel, CITIC Futures

2.5 Mineral Resources Limited

2.5.1 Onslow Iron Project

The Onslow Iron Project is developed by Mineral Resources Limited (MinRes) on behalf of the Red Hill Iron Joint Venture (RHI JV). MinRes holds a direct stake of 57% and an indirect stake of 3.3%, while Baowu Steel, AMCI and POSCO each hold equity interests of 18.7%, 10.7% and 10.3% respectively. The project adopts the “MinRes Full-Process Services + JV Partners’ Resources” model: MinRes is responsible for the project’s mining services and infrastructure construction, including designing, building and operating a 150-kilometer dedicated haul road, constructing supporting facilities at the Port of Ashburton (a transshipment terminal and 20,000 DWT transshipment vessels), and providing crushing services. The joint venture partners hold ownership of the mineral resources (the mining rights are jointly owned by MinRes, Baowu Steel, AMCI and POSCO) and will purchase the project’s output under off-take agreements.

The Onslow Iron Ore Project has total resources of 744 Mt, an average iron grade of 56.3%, and a targeted annual production capacity of 35 Mt.

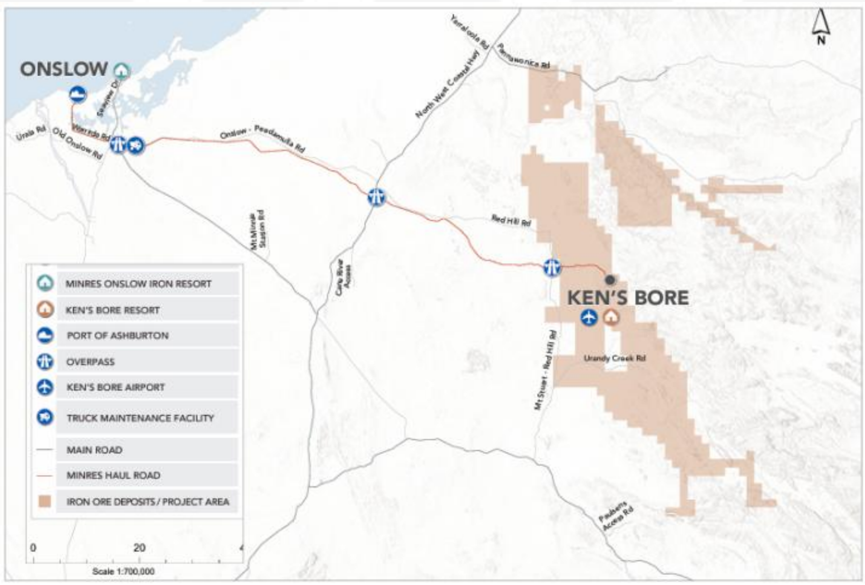
The Onslow Iron Ore Project is located in the Western Pilbara region of Western Australia. It utilizes 140 giant road trains for ore transportation, each measuring 51.6 meters in length and with a load capacity of 290 metric tons. The 150-kilometer transportation route from Kensbole Mine to the Port of Ashburton is a dedicated private haul road. There is no interaction between road trains and public vehicles, and the haul road is fenced off to prevent livestock and wildlife from accessing it. At the port, the ore is transported via enclosed transshipment vessels to waiting ships 40 kilometers offshore.

The Onslow Iron Project has a unit cost of approximately AUD 45–54 per tonne, equivalent to USD 29–37 per tonne. According to the project introduction materials on the official website, the iron ore FOB cost of the project is only AUD 45 per wet metric tonne (excluding royalty fees). The FOB cost disclosed in the quarterly report for the first quarter of Fiscal Year 2026 stands at AUD 54 per wet metric tonne.

The production capacity of the Onslow Iron Project has reached its designed level, with output continuing to ramp up. The targeted annual production capacity of 35 Mt was originally scheduled to be achieved by the September quarter of 2026. The project made its first shipment in May 2024 and has seen a smooth production ramp-up. Based on the latest shipping data, it basically reached full capacity in the third quarter of 2025, with an estimated shipment volume of 25 Mt for the full year of 2025.

The Onslow Iron Ore Project is expected to see a y/y supply increase of 7–10 Mt in 2026. The guidance target for Fiscal Year 2026 is set at 30–33 Mt. If the project maintains full-capacity operation in 2026, the y/y supply increment is projected to hit 10 Mt, with a neutral forecast of a 7-million-metric-ton y/y supply increase.

图表13：Location of the Onslow Iron Ore Project



Source: Company's website, Mysteel, CITIC Futures

2.5.2 Lamb Creek Iron Ore Project

The Lamb Creek Iron Ore Project is a key growth project under active development by Mineral Resources Limited (MinRes) in the Pilbara region of Western Australia. As an important component of MinRes' Pilbara Hub, this project serves as a crucial successor asset to sustain the Hub' s annual output of approximately 10 - 14 Mt. Its core strategic objective is to take over the production capacity of existing mines and extend the overall operational lifespan of the Hub. MinRes holds a 100% stake in the project through its wholly-owned subsidiary, Process Minerals International.

The Lamb Creek Iron Project has a targeted annual production capacity of approximately 7.5 Mt. The project's resources are hosted in layered iron ore deposits, characterized by high-grade Brockman-type mineralization with an average iron grade of 58.3%. This type of ore generally boasts superior quality and is suitable for use as a blending component to enhance the overall iron content of the final product. The ore from this project is planned to be blended with that from Iron Valley to form a standardized product named Pilbara Blend, which will then be sold to steel mill customers.

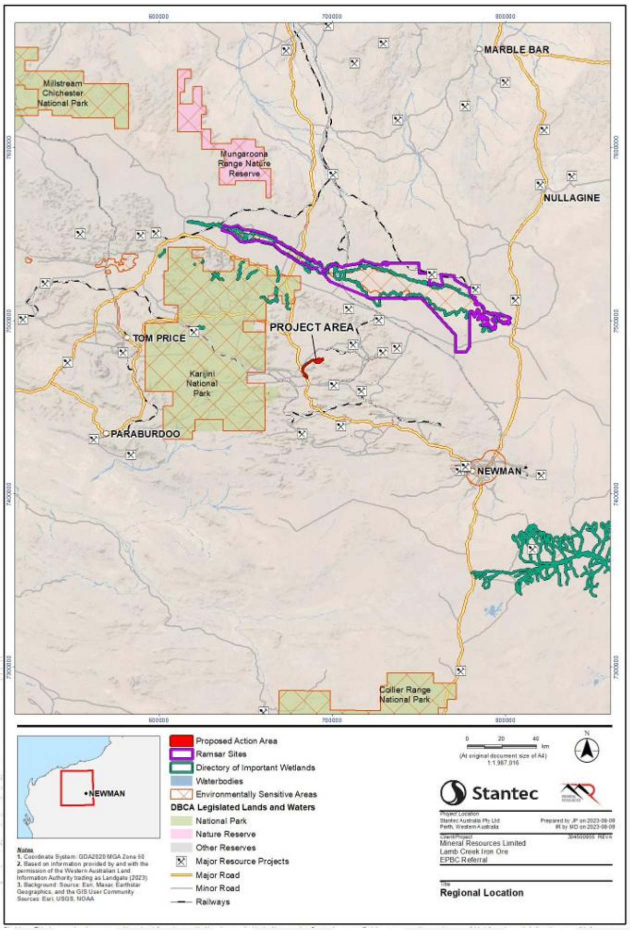
The Lamb Creek Iron Project is located in the eastern Pilbara region of Western Australia, approximately 130 kilometers northwest of Newman and 50 kilometers west of MinRes' existing Iron Valley Mine. After ore extraction, crushing and screening will be carried out on-site. The ore will then be transported from the mine via approximately 16 kilometers of dedicated haul roads using road trains. It will connect to the Great Northern Highway and be shipped over 300 kilometers to the Utah Point berth at Port Hedland for export.

The Lamb Creek Iron Project is expected to have a cash cost of AUD 75–80 per wet metric tonne, equivalent to USD 49–54 per wet metric tonne. The project will fully leverage the mature existing infrastructure of MinRes' Pilbara Hub, thereby significantly reducing capital expenditure. According to the iron ore FOB cost of the Pilbara Hub disclosed in the quarterly report for the first quarter of Fiscal Year 2026, which stood at AUD 83 per wet metric tonne, and the guidance cost for Fiscal Year 2026 of AUD 75–80 per wet metric tonne, the iron ore cost of the Lamb Creek Project is likely to be below this range.

The Lamb Creek Iron Ore Project will commence production in early 2026. The first ore extraction is scheduled for January 2026, with export expected to start in March 2026. The mine has a lifespan of 5 years, with the actual mining period spanning approximately 3 years.

As a transitional project, the Lamb Creek Iron Ore Project is not expected to contribute any supply increment in 2026. Its core role is to bridge the production gap from aging mines and extend the operational cycle of the Pilbara Hub. Although the project alone could add output volume, it may not drive an overall production increase for MinRes as a whole.

图表14：Location of the Lamb Creek Iron Ore Project



Source: Company's website, Mysteel, CITIC Futures

图表15：Schedule for the Commissioning of the Lamb Creek Iron Ore Project

Stage	Indicative Start	Indicative Finish	Anticipated Duration
Construction	**October 2025**	**February 2026**	**5 months**
Site access and mobilisation, construction camp	October 2025	February 2026	5 months
Site clearing, pre-strip (supporting infrastructure, mine, waste landforms, processing area)	October 2025	January 2026	4 months
Supporting infrastructure construction (e.g. internal roads, water supply and drainage control, permanent camp, WWTP, power generation, workshops, maintenance facilities, warehouse/offices)	October 2025	February 2026	5 months
Crushing and screening plant, stockpiles, RoM pad construction	November 2025	February 2026	4 months
Commissioning	**March 2026**	**May 2026**	**3 months**
Crushing and screening plant commissioning	March 2026	May 2026	3 months
Commence Mining Operation	**January 2026**	**December 2028**	**3 years**
Commence mining, haulage and export	January 2026		
Waste rock landform operation	January 2026	December 2028	3 years
Crushing and screening plant operations	March 2026	March 2029	3 years
Product export	March 2026	March 2029	3 years
Complete mining		**December 2028**	
Complete processing and export	December 2028	March 2029	4 months
Site decommissioning and closure	**March 2029**	**March 2034**	**5 years**
Post closure monitoring			Nominal 5 years

Source: Company's website, Mysteel, CITIC Futures

3. Brazil

3.1 Vale

3.1.1 Expansion Project of the S11D

The S11D Mining Complex is a core asset of Vale within its Northern System in northern Brazil. Wholly owned and operated by Vale, the project has a total investment of USD 2.8 billion, serving as a strategic cornerstone for the company to meet the global demand for green transition and consolidate its leading edge in high-grade iron ore supply.The expansion of the S11D Mining Complex is the centerpiece of the Northern System 240 Mtpy Project, which in turn constitutes the flagship initiative under Vale's Novo Carajás Program. Against the backdrop of the gradual depletion of resources at Serra Norte and Serra Leste, the Northern System 240 Mtpy Project is designed to fill the supply gap caused by resource exhaustion. It aims to achieve the long-term target of an annual production capacity of 240 Mt for the Northern System by integrating the resources and technologies of the Carajás Mining Complex in Pará State, Brazil.

As the largest mineralized body in Serra Sul, the S11D Deposit stretches over 28 kilometers with stable geological conditions of ore beds and proven reserves of approximately 4.24 billion metric tons. This expansion project focuses on high-grade iron ore, with an iron content maintained above 65%. Based on the historical data of the mining complex, the actual grade is highly likely to remain at a high level of around 66.7%.

The project is located in the city of Canaã dos Carajás, approximately 1,940 kilometers north of São Paulo and about 60 kilometers southwest of Vale’s largest mining complex, Serra Norte. After being crushed, the extracted iron ore is directly transported via a long-distance conveyor belt to the ore processing plant within the mining complex for beneficiation. The processed iron ore will first be shipped via a newly built 101-kilometer railway spur to Vale’s EFC main railway line, and then transported through this railway to the Maritime Terminal of Ponta da Madeira in the state of Maranhão.

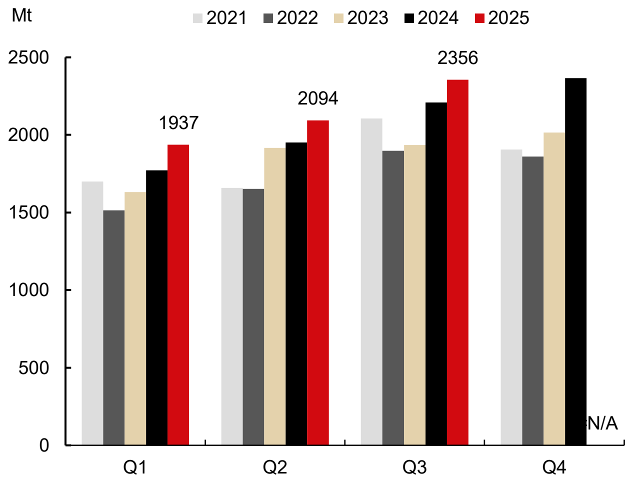
The indicative cash cost range for the S11D Mining Complex is USD 20.5–22 per metric tonne. Vale’s overall C1 cash cost for iron ore stood at USD 21.8 per metric tonne in 2024, while the S11D Mining Complex, as the company’s core low-cost mining asset, boasts a cost level below the corporate average. Vale has set the indicative C1 cash cost range for iron ore at USD 20.5–22 per metric tonne for 2025. With the advancement of the S11D expansion project and the application of technologies such as automated transportation and AI-powered ore beneficiation, its cost is expected to move closer to the lower end of this range. This expansion is a capacity ramp-up project built upon the existing mining complex, eliminating the need for large additional infrastructure investment and enabling a further reduction in the unit cost.

The capacity expansion project of the S11D Mining Complex is scheduled to commence production in the second half of 2026. The expansion of the S11D Mining Complex (also referred to as the Serra Sul 120 Project, the Serra Sul +20 Project, and the S11D Project) is an initiative to expand the ore-dressing capacity of the S11D Mine within the Northern System. It aims to raise the annual production capacity of the S11D Mining Complex from 100 Mt to 120 Mt, adding an incremental capacity of 20 Mt, and is expected to start operation in the second half of 2026. According to the quarterly report, as of July 31, 2025, the project has achieved 57% progress in terms of financial investment and 77% progress in physical construction.

图表16：S11D is part of Serra Sul

Source: Company's website, Mysteel, CITIC Futures

图表17：Quarterly Output of S11D



Source: Company's website, Mysteel, CITIC Futures

3.1.2 Capanema project

The Capanema Project is a key initiative for Vale to achieve capacity recovery and sustainable development in its Southeast System in Brazil, and it is a wholly-owned asset of Vale. The project aims to utilize an idled mine and achieve low-cost, environmentally friendly iron ore production by introducing modern processing technologies. As a core capacity expansion project of Vale's Mariana Integrated Operations Area, it plays a vital role in boosting production capacity.

The Capanema Project is expected to add an annual iron ore production capacity of approximately 15 Mt for Vale. The iron ore extracted and recovered from this project features superior quality with an iron content of 62%, and the ore produced is mainly used as a raw material for sinter.

The Capanema Project is located in the city of Ouro Preto, Minas Gerais State. As part of Vale's Mariana Integrated Operations Area, it is 80 kilometers away from Belo Horizonte and adjacent to cities such as Santa Bárbara and Itabirito. Short-distance ore transportation within the mine site is undertaken by 5 Caterpillar 789D autonomous mining trucks with a load capacity of 194 tonnes each, enabling 24/7 uninterrupted operations. A dedicated long-distance belt conveyor system has been built for the project, efficiently connecting mine extraction with subsequent rail transportation. Vale has renovated and upgraded the product storage yard and loading area of the Timbopeba railway freight station to ensure the smooth connection of ore to the main line of the Vitória-Minas Railway (EFVM). After ore loading is completed here, the ore is transported via this railway line to the Port of Tubarão in Espírito Santo State.

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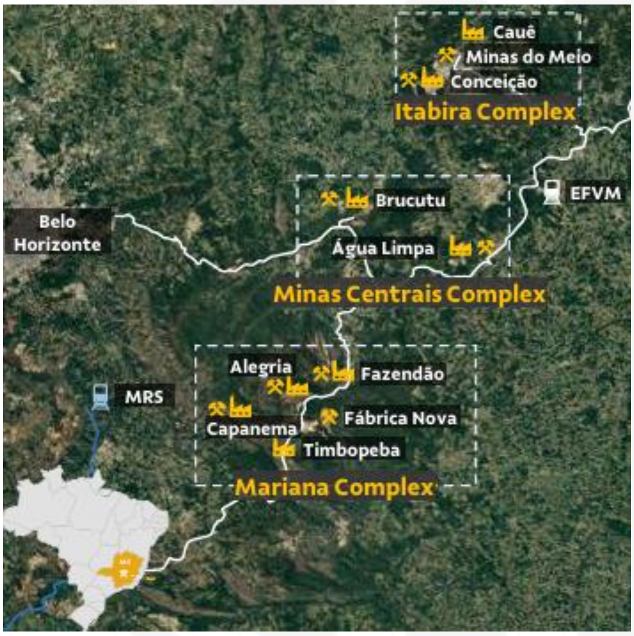
The indicative cash cost range for the Capanema Project is USD 20.5–22 per metric tonne. As a key capacity expansion project of Vale's Southeast System, the unit cost of iron ore from this project is aligned with Vale's overall 2025 C1 cash cost guidance target, i.e., USD 20.5–22 per metric tonne. Hailed as a flagship green initiative prioritized by Vale, the project adopts a variety of innovative cost-cutting technologies and models. From an internal comparative perspective, its production cost is highly likely to be at the lower end of this range or even lower.

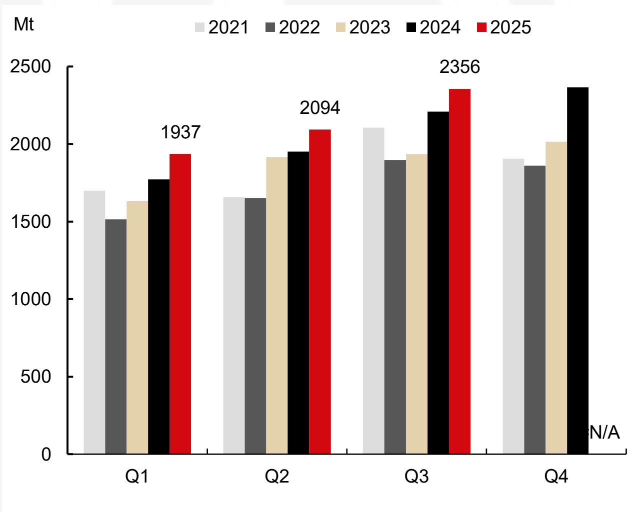
The production capacity of the Capanema Project is still in the ramp-up phase. Construction of the project kicked off in the second half of 2023, and it entered the trial operation stage in December 2024, with full-capacity operation expected to be achieved in the first quarter of 2026. A new automated ore-dressing production line is scheduled to be added in 2026, incorporating high-pressure grinding roll (HPGR) technology. This is expected to further reduce energy consumption by 15% and target an increase in iron grade to 66%. The project has seen a rapid capacity ramp-up: its output reached 600,000 metric tons in the second quarter of 2025, equivalent to an annualized capacity of 2.4 Mt; in the third quarter of 2025, output hit 2.9 Mt, with the annualized capacity reaching 11.6 Mt.

图表18：Location of the Capanema Project and the Southeast System



Source: Company's website, Mysteel, CITIC Futures

图表19：Quarterly Output of the Southeast System



Source: Company's website, Mysteel, CITIC Futures

3.1.3 Impact of Vale's New Projects

Vale's expected iron ore supply increment in 2026 is projected to be 6-10 Mt. Previously, Vale estimated its 2026 production target at 340-360 Mt, with the midpoint representing a 20 million metric ton increase compared to the 2025 production target of 325-335 Mt. According to the latest announcement, the company expects its iron ore output to reach 335 Mt in the current fiscal year, and has lowered its 2026 iron ore production guidance by 10 Mt to 335-345 Mt. This revised guidance reflects a 10 million metric ton increase at the midpoint relative to the 2025 target.

The S11D Mining Complex will commence production relatively late, while the 15 million metric ton capacity of the Capanema Project in the Southeast System is still in the ramp-up phase, which is in line with the planned schedule. However, the incremental output from these new projects will be partially offset by the production decline at aging mines. The commissioning of new production capacity is aimed at offsetting the output reduction from old mines and ensuring the achievement of the 2026 production target, meaning the actual y/y supply increment may be relatively limited. If Vale meets the lower end of its 2026 production target, its output is expected to remain flat y/y; if it hits the upper end, production is projected to rise by 10 Mt y/y. Under a neutral scenario, the y/y output growth is estimated at 6 Mt.

3.2 Samarco Iron Ore Project

The Samarco Iron Ore Project is a large-scale integrated iron ore project in Brazil jointly owned by Vale and BHP, with each holding a 50% stake. It covers the entire industrial chain including mining, ore beneficiation, transportation, pelletizing, and port export. Once suspended due to the tailings dam failure accident in 2015, the project has gradually resumed production through technological upgrading and restructuring, and is currently advancing the full recovery of its production capacity.

At the end of fiscal year 2024, BHP disclosed that the Samarco Iron Ore Project boasts mineral resources of 5.19 billion metric tons. The ore mined by the project is mainly composed of metamorphic quartz hematite and friable hematite, with an iron grade ranging from 35% to 60%. The core product of this project is iron ore pellets—a value-added processed product with a significantly elevated grade, containing approximately 67% iron.Prior to the accident, the project had an annual production capacity of 26–27 Mt, consisting of two mines (Alegria and Germano), three ore beneficiation plants, three pipelines, four pellet plants and one port. On November 5, 2015, a tailings dam of Samarco suffered a catastrophic failure, bringing the project's production to a complete halt. On December 23, 2020, Samarco gradually resumed operations, utilizing one of the three ore beneficiation plants at the Germano Integrated Mining Complex and one of the four pellet plants at the Ubu Integrated Mining Complex. This initial phase of restart achieved an annual production capacity of 7–8 Mt, accounting for 26% of Samarco's total installed capacity. In 2025, Samarco recommissioned its second ore beneficiation plant, raising its annual production capacity to approximately 14–16 Mt, which has essentially met the phased production recovery target.

The Samarco Project features the world's longest iron ore concentrate pipeline system, stretching approximately 400 kilometers from the Germano Mining Complex in Minas Gerais State to the pellet plants and port. Initially designed with an annual conveying capacity of 12 Mt, the system's actual throughput has been increased to 16 Mt through technical optimization. Subsequent efficiency improvements have been achieved by adding booster pump stations and expanding pipeline infrastructure. After being transported via the pipeline to the Ubu Complex for processing, the iron ore concentrate is directly loaded onto ships for global export at the project's owned Ubu Port. This maritime terminal can accommodate vessels with a deadweight tonnage of up to 210,000 metric tons and is equipped with a ship loader with a rated loading capacity of approximately 12,000 metric tons per hour, enabling a maximum annual loading capacity of 33 Mt.

The cash cost per tonne of iron ore at the Samarco Project is likely to be significantly higher than the $18.56 per tonne unit cash cost of BHP's Western Australian Iron Ore (WAIO). A rough estimate puts it in the range of $30 - $33 per tonne. Operational data of the Samarco Iron Ore Project is usually incorporated into the overall statistics of BHP's iron ore business. BHP's iron ore assets mainly consist of two major segments: Western Australian Iron Ore (WAIO) and Samarco Iron Ore. Its financial reports mostly disclose the overall cost of the iron ore business or the standalone cost of the WAIO segment, without separately accounting for the production costs of Samarco on a standalone basis. In fiscal year 2025, BHP reported a unit cash cost of $18.56 per tonne for WAIO iron ore. The costs of Samarco not only include regular mining and production expenses, but also incorporate substantial additional expenditures associated with production resumption, meaning its initial costs are likely to be notably higher than the aforementioned figure. According to incomplete statistics, a total of $260 million has been invested to restore initial production capacity, with earmarked funds of $580 million planned for the early closure of the Germano Dam, the last tailings dam at the Germano Mining Complex. Samarco has committed to investing $2.2 - $2.8 billion to rehabilitate all production facilities. Based on a ten-year depreciation schedule at full production capacity, this would add $11.5 - $13.8 to the unit cost per tonne. If total expenditures for environmental remediation and compensation to disaster victims are factored in, the unit cost would be even higher.

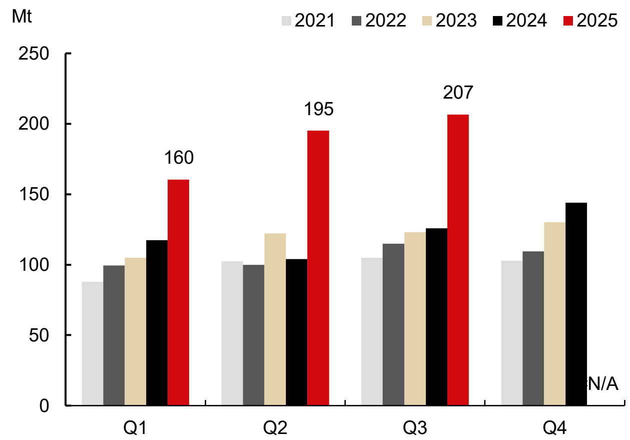
Samarco has achieved its phased production capacity recovery target, with output continuing to ramp up, and there are still plans in place for further capacity restoration in the future. In the third quarter of 2025, Samarco's 100% attributable iron ore output reached 4.14 Mt, equivalent to an annualized output of 16.56 Mt, marking a recovery to 63.7% of its total designed annual capacity of 26 Mt. The project is scheduled to reach full-capacity operation (100%) by 2028, with an annual output of 26–27 Mt.

The y/y supply increment of the Samarco Project is projected to reach 1–1.5 Mt in 2026. As of the third quarter of 2025, Samarco has achieved its short-term capacity recovery target, with its annual output for 2025 estimated at around 15 Mt. Under the scenario of full-capacity operation in 2026, the y/y supply increment will hit 1.5 Mt, while the y/y increment is expected to stand at 1 Mt based on a neutral outlook.

图表20：Location of Samarco

Source: Company's website, Mysteel, CITIC Futures

图表21：Quarterly Output of Samarco (BHP's 50% Attributable Share)



Source: Company's website, Mysteel, CITIC Futures

3.3 CSN: Pires Tailings Recovery Project

The Pires Tailings Recovery Project of CSN (Companhia Siderúrgica Nacional) is a core solid waste recycling initiative being advanced by the company in Minas Gerais State. Focused on the resource recovery and environmentally friendly disposal of tailings left over from iron ore mining, the project primarily processes iron ore tailings generated by the Pires Mining Area and its surrounding ore beneficiation plants. By upgrading technologies to replace the traditional tailings dam storage model, it achieves the dual goals of resource regeneration and environmental risk management.

CSN Mining is the second-largest iron ore exporter in Brazil, with certified iron ore reserves exceeding 2 billion metric tons and an annual production capacity of approximately 33–36 Mt at its owned mines. Combined with procured ore volumes, the company's annual iron ore exports surpass 40 Mt. CSN Mining plans to invest 13.2 billion Brazilian reals in Minas Gerais State by 2030, aiming to expand production capacity and upgrade iron ore grade. Among these initiatives, the company intends to commission the Pires Tailings Recovery Project in the first quarter of 2026, which will have an annual production capacity of 1.1 Mt and produce iron ore with a grade of 64%.

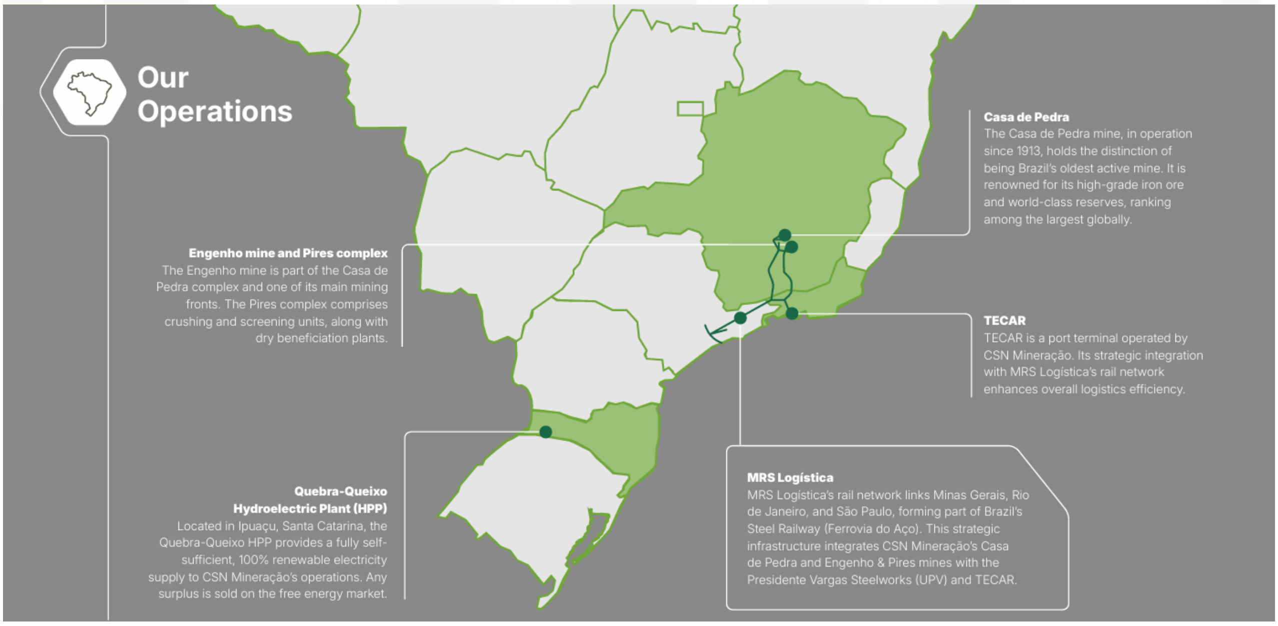
CSN holds shares in MRS Logística. In Brazil's domestic market, this railway undertakes the transportation of the majority of the company's products to serve CSN Group's owned steel mills. For export-oriented iron ore, it is transported from mine railway stations to the Port of Itaguaí in the state of Rio de Janeiro by MRS Logística.

The cash cost of the Pires Tailings Recovery Project is projected to be below $20 per metric tonne. For reference, CSN reported a C1 cash cost of $20.8 per metric tonne in the second quarter of 2025, with the disclosed C1 cash cost hovering around $20–$21 per metric tonne in the past two years. The key cost advantage of the tailings recovery project lies in the elimination of expenses associated with mining and blasting processes, which endows its cash cost with strong competitiveness and brings the actual cost to potentially below $20 per metric tonne.

The Pires Tailings Recovery Project is scheduled to be commissioned in the first quarter of 2026, with other projects to be put into operation successively in the future. CSN's capacity expansion plan consists of a series of projects, targeting an increase in annual production capacity to over 60 Mt by 2030. At its core is the construction of a new iron ore beneficiation plant, named the P15 Itabirite Beneficiation Plant, at the Casa de Pedra complex in the city of Congonhas. This plant will have an annual capacity of 16.5 Mt of high-grade pellets with an iron content of 67%, and is currently under construction, with commissioning expected in the fourth quarter of 2027.The investment package also includes initiatives for the recovery and reuse of tailings currently stored in decommissioning tailings dams. Specifically, CSN plans to commission the following projects: The Pires Tailings Recovery Project in Q1 2026, with an annual capacity of 1.1 Mt, producing iron ore with a 64% iron grade; The B4 Tailings Recovery Project in Q1 2027, with an annual capacity of 2.5 Mt, producing iron ore with a 66% iron grade;The Ultrafines Sinter Feed Recovery Project in Q4 2027, with an annual capacity of 1 Mt, producing iron ore with a 66% iron grade; The Casa de Pedra Tailings Recovery Project in Q3 2029, with an annual capacity of 2.5 Mt, producing iron ore with a 66% iron grade.

The Pires Tailings Recovery Project is expected to bring a supply increment of 0.5–1 Mt in 2026. CSN plans to commission the Pires Tailings Recovery Project, which has an annual production capacity of 1.1 Mt, in the first quarter of 2026. The project is projected to drive a y/y supply increment of around 1 Mt in 2026, with a neutral outlook for a 0.5 million metric ton y/y increase.

图表22：Location of CSN's Iron Ore Resources



Source: Company's website, Mysteel, CITIC Futures

4. Non-mainstream countries

4.1 Guinea: Simandou Project

The Simandou Iron Ore Mine, located in southeastern Guinea of West Africa, is one of the world's largest undeveloped open-pit hematite deposits with the highest iron grade. It is also Africa's largest integrated mining and infrastructure project, whose annual production capacity of 120 Mt will reshape the global iron ore supply landscape. The project is divided into the Northern Block (Blocks 1 and 2) and the Southern Block (Blocks 3 and 4), jointly developed by a consortium of international enterprises. Chinese companies collectively hold over 50% equity interest and serve as the core controlling parties of the project. The Northern Block is primarily developed by the Winning Consortium (WCS), while the Southern Block is spearheaded by Simfer.

The total investment of the Simandou Project exceeds 20 billion US dollars. It is designed to have an annual production capacity of 60 Mt for each of the northern and southern blocks, totaling 120 Mt. The proven reserves amount to approximately 4.4 billion metric tons, with the estimated total resources close to 5 billion metric tons. The average total iron grade is over 65%, coupled with low impurity content (such as sulfur and phosphorus).

The Simandou Project is located in Kérouané Prefecture, an inland area in southeastern Guinea, West Africa. Situated in a remote mountainous region, the mining area is over 600 kilometers away from Guinea's coastline and features complex terrain. To address ore transportation challenges, a 670-kilometer railway has been constructed for the project. Compagnie des Transports de Guinée (CTG) leads the development and operation of infrastructure facilities (railway and port). CTG has a shareholding structure where Winning Consortium and Simfer each hold a 42.5% stake, and the Guinean government holds the remaining 15%. In the initial phase, port facilities were built at the mouth of the Marimbaia Estuary. Ore is first transported to the port via the railway and then transferred to ocean-going vessels for export by barges. In the subsequent phase, a deep-water port is also planned to further improve transportation efficiency.

The operating cost of the Simandou Project is approximately $24.5 per metric ton. Boasting concentrated ore bodies with shallow burial depth, the project features favorable open-pit mining conditions and low extraction costs. At the end of 2023, Rio Tinto disclosed that the estimated operating cost of the project stood at around $24.5 per metric ton. However, due to the project's high upfront fixed investment and substantial infrastructure amortization expenses, its initial costs are likely to be significantly higher than this figure. As production capacity ramps up and economies of scale kick in, the cash flow cost is expected to decline gradually.

The Simandou Project is expected to contribute an iron ore supply increment of 10–20 Mt in 2026. On December 3rd, an ocean-going cargo vessel loaded with 200,000 metric tons of high-grade iron ore set sail from the Port of Marimbaia in Guinea bound for China, marking the successful shipment of the first batch of iron ore from the Simandou Project. At the end of 2025, Rio Tinto disclosed that the 2026 sales target for the Southern Block of Simandou (on a 100% basis) would be 5–10 Mt. Earlier, the company had announced plans to gradually ramp up the production capacity of the Southern Block over a 30-month period until it reaches full operation. Assuming the Northern Block progresses at the same pace, the total annual supply increment for 2026 would range from 10 to 20 Mt, slightly lower than previous market expectations. The current constraint is mainly limited transportation capacity.

图表23：Location of Simandou

Source: Company's website, Mysteel, CITIC Futures

4.2 Liberia: ArcelorMittal: Phase II Expansion Project of Nimba Iron Ore Mine

ArcelorMittal is the absolute dominant player in Liberia’s iron ore sector, operating the Nimba Iron Ore Mine. With an annual output accounting for over 80% of Liberia’s total iron ore production, it serves as the core pillar of the country’s iron ore exports. The project is jointly developed by ArcelorMittal Group (holding a 70% equity stake) and the Government of Liberia (holding a 30% equity stake), with ArcelorMittal Liberia (AML) as the operating entity. It is worth noting that the Nimba Iron Ore Mine is divided into two parts: the section located in Nimba County, Liberia (led by ArcelorMittal), and the section situated in the Nimba Mountain Range, Guinea (led by Ivanhoe Atlantic).

Since entering Liberia in 2005, the cumulative investment in the project has approached 3 billion

US dollars, among which the Phase II expansion project alone has an investment of 1.8 billion US dollars, making it one of the largest foreign-funded projects in Liberia's history. The core construction includes the Tokadeh Beneficiation Plant and its supporting facilities, which will raise the production capacity from 5 Mt to 20 Mt. ArcelorMittal holds approximately 750 Mt of iron ore resources in Nimba County, Liberia, mainly high-grade hematite, with the basic iron grade of the ore in the mining area ranging from 58% to 60%. In the second stage of the Phase II expansion, a modern beneficiation plant will be built. After magnetic separation processing, the grade of the high-grade concentrate ore for export can reach 64% to 65%.

The Nimba Iron Ore Mine is located in Nimba County, Liberia, adjacent to the Simandou Iron Ore Mine in Guinea. Its ore is mainly transported by railway to the Port of Buchanan for export. The railway line from the Yekepa Mining Area to the Port of Buchanan is the only iron ore transportation route in Liberia, stretching 243 kilometers in total. First constructed by the Liberia-American Mining Company (LAMCO) in the last century, the railway was suspended due to the civil war. Since 2005, ArcelorMittal Liberia has taken over and obtained the right to use this railway in accordance with the mineral development agreement, and has carried out rehabilitation and upgrading work on it for transporting iron ore from the Nimba Iron Ore Mine.

The reference cash flow cost of the Nimba Iron Ore Mine is approximately $30 per metric ton. The mine adopts an open-pit mining method, featuring shallow ore burial depth and low mining difficulty. According to a 2014 report by the United States Geological Survey (USGS), the production cost of iron ore by ArcelorMittal in Liberia was below $30 per metric ton. During this phase, no complex ore beneficiation process was required, resulting in a simplified production flow and cost levels that remained low within the industry. The newly built beneficiation plant under the Phase II expansion has incurred additional expenditures related to the operation of relevant equipment, energy consumption, and ore processing stages. This may lead to a significant short-term increase in the unit operating cost per ton of ore. However, with the substantial ramp-up of production capacity, the unit operating cost is expected to be gradually diluted in the future.

The Phase II expansion project of the Nimba Iron Ore Mine has been commissioned, with its output ramping up gradually. The iron ore beneficiation plant, a core component of ArcelorMittal's $1.8 billion Phase II expansion project, was officially completed and commissioned in June 2025 by ArcelorMittal Liberia. This has boosted the mine's annual production capacity sharply from 5 Mt to 20 Mt. For the long-term plan (post-2026), the company will explore the feasibility of further raising the annual production capacity to 30 Mt, continuously optimize the industrial chain and supporting infrastructure. Meanwhile, it will also explore the production of direct reduced iron (DRI)-grade concentrate to align with the low-carbon transformation trend of the global steel industry and enhance the competitiveness of its products in the high-end market.

The expansion project is expected to drive a y/y iron ore supply increment of 5–9 Mt in Liberia. An additional annual production capacity of 15 Mt was commissioned in June 2025, with output ramping up gradually. Liberia's iron ore shipments are projected to register a y/y increase of 9 Mt in 2026, with a neutral outlook for a 5 million metric ton increment.

图表24：Location of Nimba Iron Ore Mine

Source: Company's website, Mysteel, CITIC Futures

4.3 South Africa: Palabora: Phase II Deep Development Project

The iron ore project operated by Palabora Mining Company (PMC) is the only large-scale copper-iron associated ore project in South Africa controlled by a Chinese enterprise (HBIS Resources). Currently, this project is a core asset of HBIS Resources and also a rare global model for the development of polymetallic associated ores. The project can stably supply approximately 8 Mt of high-grade iron ore to the Chinese market annually. Leveraging its unique resource endowments, low-cost operational advantages and a steadfast focus on the Chinese market, it has become South Africa’ s third-largest iron ore producer and occupies a special position in the global supply of non-mainstream iron ore.

This project is a rare global polymetallic associated ore deposit of copper, iron, phosphorus, and vermiculite. Its iron ore resources are mainly divided into two categories:① Tailings stockpile resources: As of mid-2025, the magnetite stockpile volume stood at approximately 120 Mt with an average grade of 58%. After magnetic separation processing prior to sales, the iron content can be further upgraded to a maximum of 65%, whose quality meets the requirements of blast furnace steelmaking. The existing stockpile resources alone can support stable production for over 10 years.② Deep-seated primary resources: The proven total magnetite reserves exceed 1 billion metric tons, associated with copper ore (with an average copper grade of 0.68%–0.8%). The main shaft of the deep mining area reaches a depth of 1,200 meters, providing resource guarantee for long-term production capacity release.The original open-pit mine and the Phase I underground mine were designed with an annual production capacity of 10 Mt. The Phase I project was on the verge of closure in 2023. HBIS Resources maintained stable output through reprocessing of tailings, achieving an actual output of approximately 7.97 Mt in 2024, accounting for 13.8% of the combined output of South Africa’s three major iron ore producers.The Phase II capacity expansion project plans to invest RMB 6 billion to advance the deep mining Phase II project, with the designed annual production capacity increased to 11 Mt. Upon the expected commissioning in 2026, it will add 3 Mt of magnetite annually and extend the mine’s service life by 15 years.The core product is high-grade magnetite concentrate (with a total iron grade of 64.5%–65%) featuring low impurity content, which is suitable for the blast furnace steelmaking needs of high-end steel enterprises in China, Europe, and other regions.

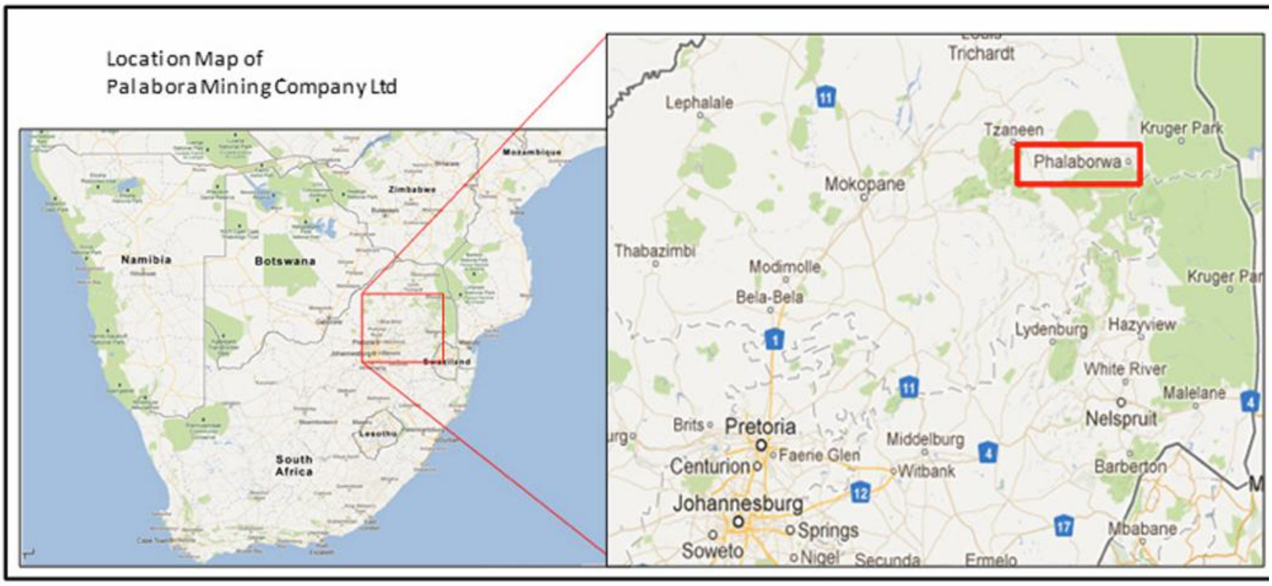
The project is located in the town of Phalaborwa, Limpopo Province, South Africa. Ores can be transported to ports on the eastern coast of South Africa via railway or highway. Adjacent to the National Route R40, the project enables quick connectivity to major industrial cities and overseas ports of South Africa. It is approximately 800 kilometers away from the Port of Richards Bay and 400 kilometers from the Port of Maputo in Mozambique. Magnetite ores are transported from the Palabora Mining Area to the Port of Maputo, Mozambique via on-site loading railways (or short-haul road transportation), while the Port of Richards Bay serves as an alternative route (to address congestion or maintenance at the Port of Maputo). The sales channels are highly concentrated and stable: the vast majority of products are under the unified underwriting of smart union resources (hong kong), with priority supply to HBIS Group internally and other major steel producers in China, making it an important alternative source for China's iron ore imports.

The cash flow cost of the project's iron ore is significantly lower than that of global mainstream iron ore projects. The magnetite of the company is a by-product separated during the copper ore processing. Its grade can be upgraded to a maximum of 65% through simple magnetic separation alone, without bearing additional mining costs, and only incurring refining and processing expenses. Compared with the multi-stage processes of crushing, grinding and flotation in traditional ore beneficiation, the labor and equipment maintenance costs are substantially reduced, resulting in the comprehensive mining and beneficiation cost being far below that of global mainstream iron ore projects. After the depletion of open-pit mine resources, the shift to deep underground mining may lead to a certain cost increase. Tailings resources are directly recycled from surface stockyards, with no extra mining costs required. The tailings reprocessing model eliminates huge upfront investments such as exploration and mining site construction. The fixed cost per unit of product is amortized through continuous production, resulting in a gross profit margin rarely seen among global iron ore projects, with transportation costs accounting for the largest share of expenditures.

The project is scheduled to be completed and reach full production capacity in July 2026. Upon achieving full production, it will add an annual output of 3 Mt of magnetite. However, its railway transportation capacity is constrained by infrastructure limitations, and shipment efficiency may affect the pace of production capacity release.

The Phase II deep mining project is expected to bring a supply increment of 1–1.5 Mt in 2026. After its completion and full production in July 2026, it will add an annual output of 3 Mt of magnetite and extend the mine’s service life by 15 years. The project is projected to contribute a supply-demand increment of 1.5 Mt in 2026, with a neutral outlook for a y/y increment of 1 Mt.

图表25：Location of Palabora



Source: Company's website, Mysteel, CITIC Futures

4.4 India

4.4.1 KIOCL: Devadari Iron Ore Project

The Devadari Iron Ore Project is a core revitalization project of Kudremukh Iron Ore Company Limited (KIOCL). As KIOCL's first proprietary and self-operated mine since the closure of the Kudremukh Mine in 2006, it undertakes the strategic mission of enabling the enterprise to break free from raw material dependence and rebuild an independent supply system. As a state-owned iron ore export and pellet production enterprise directly under the Ministry of Steel of India, KIOCL previously relied primarily on iron ore supplies from National Mineral Development Corporation (NMDC) to feed its 3.5 Mt per annum pellet plant in Mangaluru, which resulted in high raw material costs and insufficient supply stability. The implementation of the Devadari Project will achieve raw material self-sufficiency, reduce reliance on third-party suppliers, and at the same time support the raw material demand of KIOCL's pellet plant in Mangaluru as well as its subsequent ore beneficiation and pelletizing businesses, thereby enhancing the company's competitiveness in India's steel industry chain.

The Devadari Iron Ore Mine is located in Sandur taluk, Bellary district, Karnataka, and forms part of the iron ore belt of the Deccan Plateau. This area is rich in iron ore resources, with an ore grade ranging from approximately 55% to 65%. Featuring shallow ore burial depth, it is suitable for open-pit mining. The total area of the mining concession covers 388 hectares, with a lease term of up to 50 years. The lease officially came into effect after the agreement was signed with the

Department of Mines and Geology of the Government of Karnataka on January 2, 2023, and completed registration on January 18 of the same year. The project advances investment in phases. The Phase I project, evaluated by the Public Investment Board of India, has an estimated total investment of 8.8246 billion Indian rupees (equivalent to approximately 106 million US dollars). Subsequently, as the production capacity is upgraded to 2 Mt per annum, the construction of supporting beneficiation plants will entail substantial additional investment, and the total investment of the project is expected to further increase.

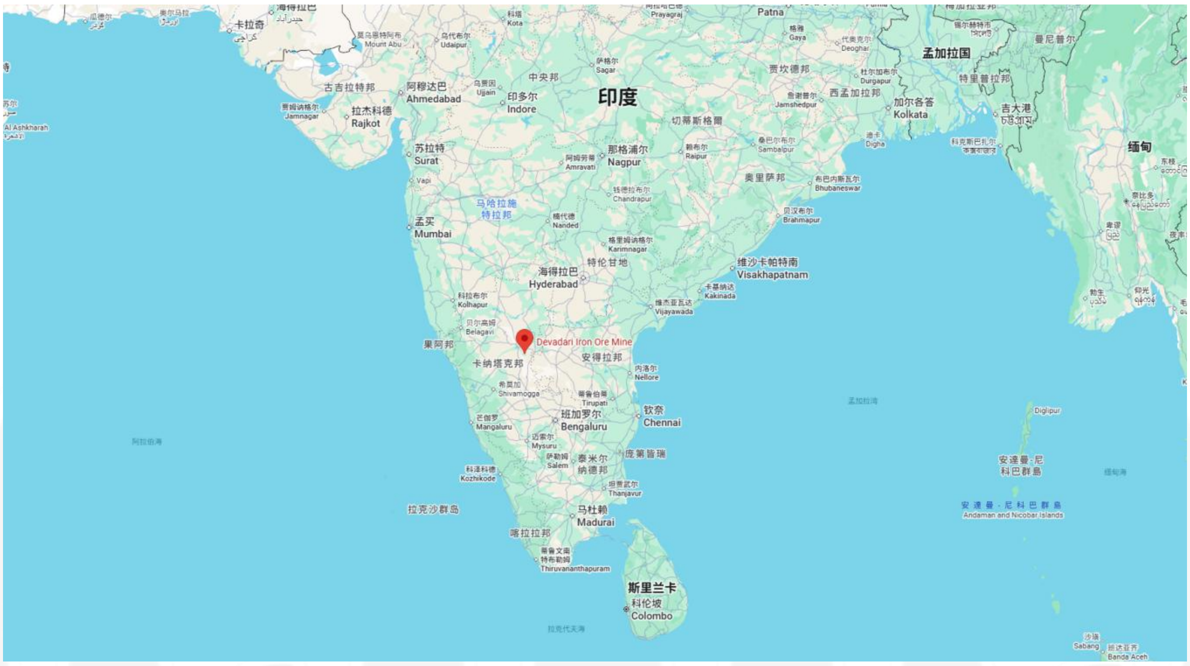
The Devadari Iron Ore Project is strategically located in the Devadari mountainous area of Sandur taluk, Bellary District, Karnataka, India. Run-of-mine ore extracted via open-pit mining will be transported by road to the pellet plant in Mangaluru (approximately 150 kilometers away). Subsequently, a supporting beneficiation plant with an annual capacity of 2 Mt will be constructed to process the run-of-mine ore into iron concentrate with a grade of 65%. The final products will be supplied to domestic steel mills via Karnataka’ s highway network or shipped to ports for export.

The Devadari Iron Ore Project is primarily intended for domestic production. There is currently no available information regarding the project’ s costs, though its cost level ranks in the middle tier among global iron ore mines. The iron ore produced is mainly supplied to the domestic market, with minimal volumes allocated for export.

The production capacity of the Devadari Iron Ore Project is planned to rise from 500,000 metric tons to 1,000,000 metric tons in 2026. The project adopts a capacity ramp-up strategy of "gradual scaling and phased achievement of full production", with clear targets for each phase: 2024–2025 fiscal year: Pilot production phase. Initial mining operations will be launched, with a target annual output of 300,000 metric tons of iron ore. This phase will primarily verify mining processes, equipment operation efficiency, and supply chain coordination. 2025–2026 fiscal year: Capacity expansion phase. Annual output will be increased to 500,000 metric tons. Mining processes will be progressively optimized, and the operational area will be expanded. 2026–2027 fiscal year: Large-scale production phase. Annual output will exceed 1,000,000 metric tons, and optimization and upgrading of major production facilities will be completed. 2027–2028 fiscal year: Full-capacity operation phase. Subject to obtaining all environmental permits, the project will achieve its designed full production capacity of 2,000,000 metric tons per annum. A supporting beneficiation plant will be built simultaneously. Gangue minerals will be removed through beneficiation processes to upgrade ore grade and realize product value increment. As a core supporting facility, the construction of the beneficiation plant will be advanced based on the Detailed Project Report (DPR) compiled using comprehensive exploration data. It is scheduled to be completed in tandem with the full-capacity production target. Upon completion, the plant will have an ore processing capacity of 2,000,000 metric tons per annum, enabling integrated operation of mining and processing.

The supply increment of the Devadari Iron Ore Project will be relatively modest, with an expected contribution of around 500,000 metric tons in 2026. It should be noted that the project is located in an ecologically sensitive area, and mining activities may trigger multiple environmental impacts. Permits related to forests and wildlife are currently in progress, along with the filing of an ecological assessment covering an area of 973 acres (including an 840-acre area to be submerged by the dam). KIOCL has committed to mitigating the impacts through ecological compensation, investment in environmental protection facilities and other measures, but the effectiveness of these measures still requires long-term monitoring and verification.

图表26：Location of the Devadari Iron Ore Project



Source: Company's website, Mysteel, CITIC Futures

4.4.2 LMEL: Surjagarh Iron Project

The Surjagarh Iron Ore Mine is one of the largest-scale and highest-potential iron ore projects in Maharashtra, India, with its mining lease held by Lloyds Metals & Energy Limited (LMEL). In 2020, LMEL and Thriveni Earthmovers Pvt Ltd (TEPL) jointly funded the establishment of a joint venture named Thriveni Lloyds Mining, which is specifically responsible for the mining, operation and other related work of the Surjagarh Iron Ore Mine. Among the partners, TEPL holds a 60% equity stake in the joint venture, while LMEL holds only 40%.

The project’s mining lease is valid until May 2057. The total geological resources of the mine amount to 857 Mt, including 156 Mt of hematite reserves which are classified as Direct Shipping Ore (DSO). With high grade, this ore can meet the requirements of steel production without complex beneficiation processes. The reserves of Banded Hematite Quartzite (BHQ) stand at 701 Mt, featuring a relatively low iron grade (30% ‑ 35%). However, its grade can be upgraded to 67% through beneficiation technologies, rendering it of extremely high development and utilization value. The company plans to invest a total of 600 ‑ 650 billion Indian rupees (equivalent to approximately

7.2 – 7.8 billion US dollars) to expand mine production capacity and extend the downstream industrial chain.

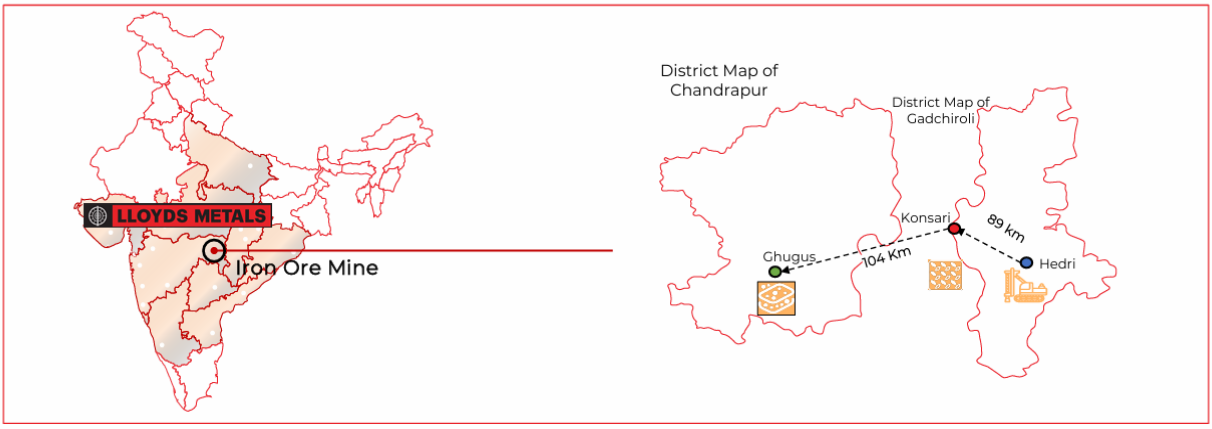
The Surjagarh Iron Ore Mine is located near Surjagarh Village in Etapalli Tehsil, Gadchiroli District, Maharashtra, India. It sits in the region with the most abundant iron ore resources in the state and is situated in central India. The mineral development infrastructure in the surrounding area is gradually improving, and it is equidistant from most steel mills, providing geographical advantages for large-scale mining and direct delivery to end users. Ore from the project is transported directly to end users via railway or pipeline. In 2025, LMEL commissioned an 85-kilometer-long iron ore slurry pipeline with an annual transportation capacity of 10 Mt. For the second phase, the company plans to extend the pipeline to a length of 195 kilometers.

The estimated cash cost of the Surjagarh Iron Ore Project is around $40 – $50 per metric ton. Derived from the company-disclosed realisation per ton minus EBITDA per ton, this figure represents the operating cash cost, which covers variable/operating expenses such as mining, processing, transportation, royalties and variable taxes. It excludes depreciation/amortisation, interest, one-off project costs, and the amortisation of capital expenditures.

Recently, the production capacity of the Surjagarh Iron Ore Project has been expanded from 10 Mt to 26 Mt. The mine had an initial operational capacity of 3 Mt per annum of run-of-mine ore (ROM). It completed the first-phase expansion in Fiscal Year 2023, raising its annual ROM capacity from 3 Mt to 10 Mt. A supporting crushing and screening plant was built simultaneously, with an annual total processing capacity of 11.76896 Mt (including 10 Mt of ROM and 1.7692 Mt of tailings). The project maintained stable production at this capacity level throughout Fiscal Year 2024 (FY24) and Fiscal Year 2025 (FY25). The second-phase expansion is being advanced gradually in Fiscal Year 2026. In June 2025, the Expert Appraisal Committee (EAC) of India’s Ministry of Environment, Forests and Climate Change (MOEFCC) approved the capacity increase to 26 Mt per annum, and the project is currently in the ramp-up phase. Subsequently, the company obtained key environmental permits again, with the final expansion target set at 55 Mt per annum. This capacity will integrate a beneficiation capacity of 45 Mt of Banded Hematite Quartzite (BHQ). Upon the commissioning of the beneficiation plant, the mine will gradually shift from direct sales of hematite to sales of high-quality beneficiated ore, forming a large-scale and diversified product supply system. The 2024 – 25 fiscal report mentioned that the 55 million metric ton capacity target will be achieved within 4 – 5 years. After expanding to 55 Mt per annum, the project will become India’s largest single iron ore mine in terms of production capacity, significantly boosting India’s iron ore self-sufficiency rate and reducing its reliance on imported iron ore.

The Surjagarh Iron Ore Mine is expected to contribute an output increment of 5 – 10 Mt in 2026. LMEL’s Surjagarh Iron Ore Mine currently has an annual production capacity of 10 Mt, with an output of 10 Mt in Fiscal Year 2025. In June 2025, it obtained approval to raise its annual production capacity to 26 Mt. Production is planned to gradually ramp up to full capacity starting from Fiscal Year 2026, with an expected output of 20 – 22 Mt. However, the actual output in the first two quarters of Fiscal Year 2026 reached 7.4 Mt, indicating that the pace of production capacity release may fall short of expectations. The mine is projected to contribute an increment of 10 Mt in 2026, with a neutral outlook of 5 Mt.

图表27：Location of the Surjagarh Iron Ore Mine



Source: Company's website, Mysteel, CITIC Futures

4.4.3 Series of Auction-based Resumption Projects in Goa

The situation regarding the "newly added iron ore production capacity" in Goa, India is rather special. Unlike the "greenfield projects" in Simandou or Australia that involve developing entirely new and untapped resources, the so-called "newly added production capacity" in Goa is actually achieved by restarting the capacity of long-idled "old mines" through government auctions. Since the Supreme Court of India revoked all mining leases in 2018, leading to a complete suspension of mining operations, Goa has been reallocating mining rights via public auctions since the end of 2022. Between 2022 and 2024, Goa held three rounds of auctions for 12 mining blocks.

The Supreme Court of India has set a cap on the total annual iron ore output of Goa, which currently stands at approximately 20 Mt per annum. Goa, India, once suffered severe environmental problems and corruption scandals triggered by large-scale illegal mining. In September 2012, responding to relevant lawsuits, the Supreme Court of India imposed a ban on iron ore mining across Goa. On April 21, 2014, in the judgment lifting this ban, the Supreme Court explicitly set the annual iron ore mining cap of Goa at 20 Mt, and simultaneously mandated the strict enforcement of this output standard pending the submission of relevant reports by the expert committee. Prior to the implementation of the ban, the annual export volume of iron ore from Goa once exceeded 50 Mt. Between 2022 and 2024, the Goa state government conducted three rounds of auctions for iron ore mining blocks in December 2022, March 2023, and September 2024 respectively, putting a total of 12 mining blocks up for auction. In the process of restarting the mining industry, the state government re-emphasized that the combined output of all operational mining blocks must be kept within the 20 million metric ton threshold set by the Supreme Court, and has initiated the formulation of a mechanism to allocate specific production quotas to the auctioned mining blocks.

The estimated iron ore reserves of Goa amount to 1 billion metric tons, with major mining areas concentrated in the mountainous regions of North Goa and South Goa, including Bicholim, Sanguem and Quepem. The ore is dominated by low-grade hematite, typically with an iron content ranging from 50% to 58%. The ore bodies are amenable to mining and are located close to ports, enabling exports via the local Mormugao Port, which confers significant geographical advantages in terms of transportation and export.

Goa’s iron ore is expected to contribute an output increment of 2–4 Mt in 2026. Among the 12 auctioned mining blocks, only Vedanta’s Bicholim-Mulgao block has been put into operation, with a production capacity of 3 Mt. It achieved an actual output of 0.9 Mt in the 2024–25 fiscal year, and it remains uncertain whether the block has reached full capacity. Salgaocar Shipping’s Shirigao-Mayem block has commenced operations, boasting a capacity of 1 Mt, though no output data is available for the time being. Vedanta’s Cudnem Block VII, JSW Steel’s Cudnem Block VI and Kai International’s Tivim-Pirna (Block VIII) have obtained environmental approvals and are expected to start operations in 2026, with respective capacities of 0.5 Mt, 0.05 Mt and 0.33 Mt. Assuming all the supply increments are released in 2026, it is projected to drive a y/y output growth of 3.98 Mt; under a neutral outlook, the y/y supply increment will stand at 2 Mt.

图表28：Location of Goa

Source: Company's website, Mysteel, CITIC Futures

图表29：Overview of Auctioned Projects in Goa

Auction Round	Mining Block Name	Successful Bidder	Reserves	Capacity	Progress
The First Round of Auctions (December 2022)	Bicholim-Mulgao	Vedanta Ltd.	Over 84 Mt	3 Mt	Commissioned
	Shirigao-Mayem	Salgaocar Shipping	About 24 Mt	1 Mt	Commissioned
	Monte-de-Shirigao	Rajaram Bandekar (RMB)	About 9.15 Mt	0.5 Mt	Under Permit Approval
	Kalay	Sociedade De Fomento Industries Private Limited (SFIPL)	About 16.73 Mt	1 Mt	Under Permit Approval
The Second Round of Auctions (March 2023)	Cudnem VII	Vedanta Ltd.	About 8.3 Mt	0.5 Mt	It is in the stages of signing the Mine Development and Production Agreement (MDPA), site preparation and commissioning of operations.
	Cudnem VI	JSW Steel	About 9.77 Mt	0.05 Mt	Under development; environmental permit has been obtained to date.
	Surla – Sonshi (Block IX)	JSW Steel	About 65.73 Mt	1.1 Mt	The project has entered the formal environmental approval phase.
	Adval Pale – Tivim	Fomento Resources	About 3.82 Mt	0.3-0.5 Mt	The environmental approval process is underway.
	Tivim – Pirna (Block VIII)	Kai International	About 8 Mt	0.33 Mt	Under development; environmental permit has been obtained to date.
The Third Round of Auctions (September 2024)	Onda (Block X)	Rebidding	About 5.6 Mt	0.5 Mt	Under Permit Approval
	Curpem 和 Sulcorna (Block XI)	Rebidding	About 8.7-9 Mt	0.6-0.8 Mt	Public Hearing / ToR / EIA Review Phase
	Codli (Block XII)	JSW Steel	About 23-24 Mt	3 Mt	The environmental approval process is progressing.

Source: Company's website, Mysteel, CITIC Futures

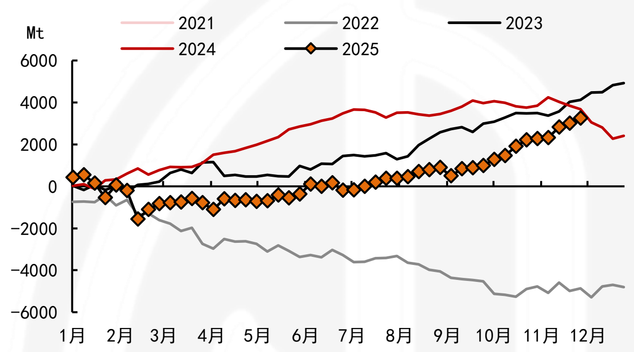
Risks: This paper compiles, to the best of its ability, the new capacity projects that are likely to drive supply increments in 2026. The impact of mine replacement has been factored in for some companies, though the completeness of the statistics cannot be guaranteed. The commissioning schedule is subjectively inferred based on the information disclosed by the companies. The cash/operating costs of some projects are rough estimates and are for reference only.

附录：中文版

1. 摘要

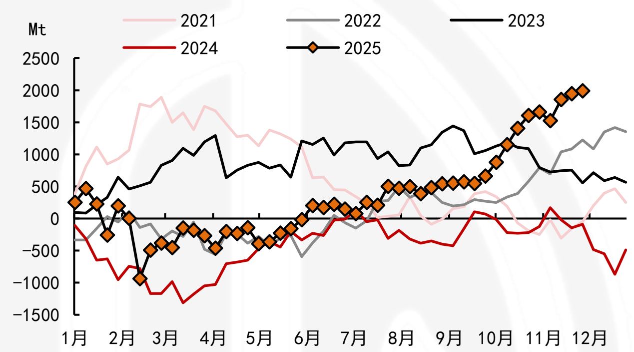
2025 年全球铁矿供给增量不及预期，上半年受天气扰动和投产不及预期的影响，发运同比减量，下半年才逐渐回升，截至 12 月初全球总发运同比增加 3200 万吨左右。发运增量主要来自澳巴非主流地区，特别是昂斯洛项目产量释放明显，四大矿山仅 FMG 铁桥项目释放较多增量，非主流国家发运同比减量。

图表1：全球总发运:累计:同差



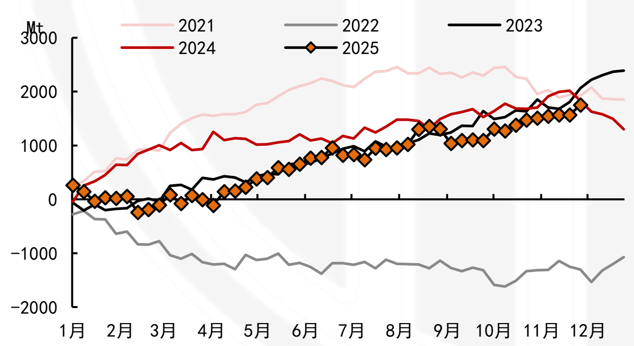
Source: Company's website, Mysteel, CITIC Futures

图表2：澳洲总发运:累计:同差



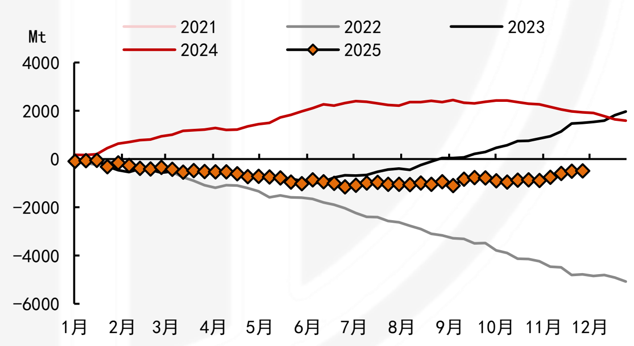
Source: Company's website, Mysteel, CITIC Futures

图表3：巴西发运量:累计:同差



Source: Company's website, Mysteel, CITIC Futures

图表4：非主流国家总发运:累计:同差



Source: Company's website, Mysteel, CITIC Futures

较 2025 年初市场预期，铁矿产能投放节奏明显延后，但大概率仍将陆续释放。本文梳理了海外矿山预计将在 2026 年释放供应增量的 16 个项目，帮助投资者明晰铁矿石 2026 年的整体供应趋势。

海外潜在新增产能较多，全球铁矿供给格局趋于宽松。据不完全统计，2026 年海外仍有较多新增铁矿产能投放，若这些产能正常投放，2026 年海外矿山将同比增加 8500 万吨。但产能投放初期可能有较多意外扰动，2025 年初市场普遍预期较多产能投放，而实际供给增量不及预期，考虑产能投放不及预期及矿山产能替代的情况，相对中性预期下，海外供给同比增量仍达 4750 万吨，供给较为宽松。

新增产能成本处于低位水平，相对于国内矿山和高成本非主流矿山仍具有明显优势。根据项目披露或粗略估算，本文梳理的新增产能中，现金成本基本位于20-50美元/吨附近，全部低于60美元/吨，位于全球50%的成本区间以内，具有较为明显的成本优势。

图表5：外铁矿供应增量统计

	Project's newly added capacity	Capacity launch progress(plan)	Optimistic Scenario: Supply Increment	Neutral Scenario: Supply Increment	C1 Cash Cost
Australia		-	**2750**	**1650**	
FMG: Iron Bridge Magnetite Project	Newly added 22 Mt	First production achieved in 2023, with capacity ramping up	500	300	33-38美元/湿吨
Rio: Western Range Iron Ore Project	Newly added 25 Mt	Already commissioned, with production volume ramping up.	500	250	23-24.5美元/湿吨
Fenix: Weld Range Project	Newly added 8.6 Mt; phased capacity expansion from 1.4 Mt to 10 Mt	Initial shipment of the sub-mine commenced in August 2025; other mines are undergoing capacity expansion.	250	150	45-53美元/湿吨
Hancock: McPhee Creek Project	Newly added 9.5-9.7 Mt	Target: 1st ore production in FY25 - 26; No commissioning info available yet.	500	250	现金成本预计较低
MinRes: Onslow Iron Project	Newly added 35 Mt	Annualized capacity: Fully operational	1000	700	29-37美元/吨
MinRes: Lamb Creek Project	Newly added 7.5 Mt	Start production: Early 2026 | Role: Capacity replacement transitional project			49-54美元/湿吨
Brazilian			**1250**	**750**	
Vale: Serra Sul+20 Project	Newly added 20 Mt; phased capacity expansion from 100 Mt to 120 Mt	Commissioning: H2 2026 | Purpose: Replace old mine	400	200	20.5-22美元/吨
Vale: Capanema Project	Newly added 15 Mt	In operation, with capacity still ramping up.	600	400	20.5-22美元/吨
Samarco Project	Newly added 7-8 Mt; phased capacity expansion from 7-8 Mt to 14-16 Mt	Phased capacity recovery target achieved; production ramping up; further recovery planned.	150	100	粗略测算可能在30-33美元/吨
CSN: Pires Tailings Recovery Project	Newly added 1.1 Mt	Commissioning: Q1 2026	100	50	预计低于20美元/吨
Guinea			**2000**	**1000**	
Simandou Project	Newly added 120 Mt	First shipment delivered; currently in the ramp-up phase.	2000	1000	预计高于24.5美元/吨
Liberia			**900**	**500**	
ArcelorMittal: Phase II Expansion Project of Nimba Iron Ore Mine	Newly added 15 Mt; phased capacity expansion from 5 Mt to 20 Mt	Production started in June 2025; capacity is ramping up.	900	500	约30美元/吨
South Africa			**150**	**100**	
Palabora: Phase II Deep Development Project	Newly added 3 Mt,Capacity Expansion Project	Commissioning: Jul 2026	150	100	低于全球主流铁矿项目
India			**1450**	**750**	
KIOCL: Devadari Project	Newly added 1.5 Mt; phased capacity expansion from 0.5 Mt to 2 Mt	Production volume surpass 1 Mt in FY2026 - 2027	50	50	
LMEL: Surjagarh Project	Newly added 16 Mt; phased capacity expansion from 10 Mt to 26 Mt	ramping up	1000	500	估算在40-50美元/吨附近
A series of auction-based resumption projects in Goa	No more than 20 Mt	A few projects operational	400	200	
Total	-	-	**8500**	**4750**	

Source: Company's website, Mysteel, CITIC Futures

2. 澳大利亚

2.1 FMG 铁桥项目

FMG 铁桥项目（Iron Bridge Magnetite Project）是澳大利亚矿业巨头 Fortescue Metals Group（简称 FMG，中文名“福德士河”）开发的高品位磁铁矿项目，位于西澳大利亚州皮尔巴拉地区，是全球少数大型高品位磁铁矿开发项目之一，FMG Magnetite Pty Ltd（FMG 磁铁私人有限公司）持股 69%，Formosa Steel IB Pty Ltd（台塑钢铁关联公司）持股 31%。FMG Magnetite Pty Ltd 是 FMG Iron Bridge Ltd (FMG 铁桥有限公司) 的全资子公司，FMG Iron Bridge Ltd 由 FMG 持股 88%，宝钢资源国际有限公司持股 12%。

铁桥项目总投资约 39 亿美元，探明储量 7.16 亿吨（矿石品位 67%），总矿产资源量 54.48 亿吨（矿石品位 30.4%），预计年产 2200 万吨含铁量为 67%的高品位磁铁精矿。

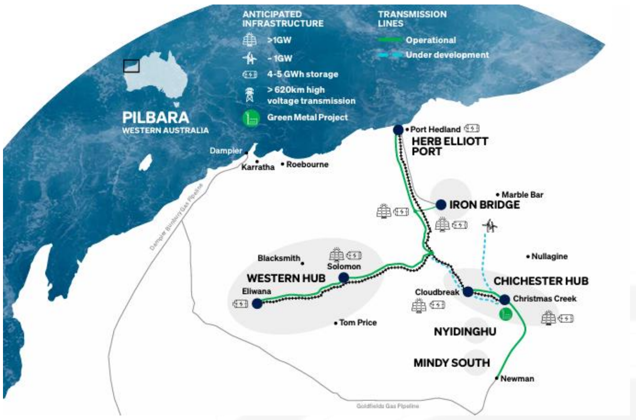
铁桥项目产品运输方便。铁桥项目位于西澳大利亚州的皮尔巴拉地区，在黑德兰港以南 145 公里处。项目采用管道运输，建有 135 公里的矿浆和回水管道以及 179 公里的高压玻璃纤维增强塑料（GRP）原水管道，将湿精矿从矿山运输到黑德兰港，在港口进行脱水和物料处理，转化为高品位磁铁矿产品。

铁桥项目现金成本约 33~38 美元/湿吨。从运营成本来看，早期钢联曾报道，铁桥项目采矿年限内该项目现金成本为 33~38 美元/湿吨，其中包括 FMG 港口及电力服务的费用。相比于国内铁矿山而言，成本优势也十分明显。

铁桥项目产能爬坡持续。2023 年 8 月铁桥项目转为运营生产，2023 年 9 月首次装运高品位（67%铁）磁铁矿产品。铁桥分阶段产能提升，根据 FMG2025 年 5 月公告最新规划，Fortescue 预计铁桥项目在 2026 财年（自然年 2025 年 7 月-2026 年 6 月）的年发货量为 1000 万至 1200 万吨；到 2027 财年（自然年 2026 年 7 月-2027 年 6 月）这一数字将提升至 1600 万至 2000 万吨。最终目标是该项目在 2028 财年（自然年 2027 年 7 月-2028 年 6 月）实现 2200 万吨的满负荷产能，较此前设定的 2025 年 9 月全面达产目标推迟近三年。2025Q3 铁桥项目加工量为 230 万吨，年化 920 万吨。

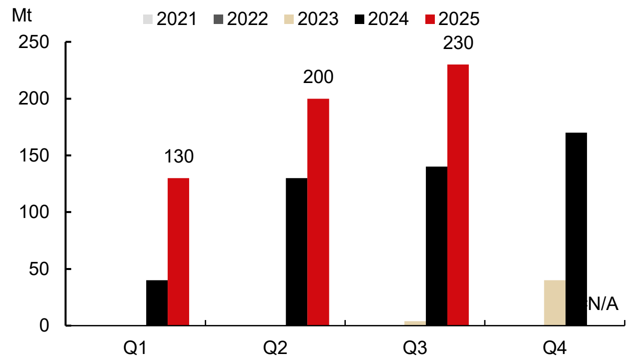
在铁桥项目支持下，2026 年 FMG 供应增量约 300-500 万吨。FMG 在 FY26 总出货量的指导为 1.95-2.05 亿吨较 FY25 财年增加 500 万吨，其中包括来自铁桥的 1000-1200 万吨（100%基础），出货指导增量主要来自于铁桥项目。25 财年铁桥项目加工量 640 万吨，假设季度年化产量线性增长至指导目标高位，预计 2026 年发运同比增量 500 万吨，若季度年化产量线性增长至指导目标低位，预计 2026 年发运同比增量 300 万吨。

图表6：铁桥项目位置



Source: Company's website, Mysteel, CITIC Futures

图表7：铁桥项目季度加工量



Source: Company's website, Mysteel, CITIC Futures

2.2 西坡项目

西坡项目是力拓在皮尔巴拉地区推进的矿山替代计划的核心项目之一，旨在维持其“皮尔巴拉混合矿（Pilbara Blend）”的稳定产能，力拓持有54%股权，中国宝武占比46%。另一方面，“西坡”项目事实上是对宝瑞吉（东坡）项目的延续，东坡项目是由原宝钢集团与力拓子公司哈默斯利铁矿有限公司合资组建，旨在开发哈默斯利山脉东坡的矿藏，产量上限为2亿吨，于2020年10月已经完成该上限。

西坡项目总投资达20亿美元，由力拓集团（Rio Tinto）与中国宝武钢铁集团（Baowu Steel Group）合资开发。项目产能2500万吨，预计可生产2.75亿吨铁矿石，平均品位约62%，服务年限达20年（按2500万吨/年计算，服务年限约11年，但力拓曾表示可根据市场需求延长）。宝武和力拓还达成一项铁矿石采购协议，宝武将按照权益比例采购力拓总量1.26亿吨的铁矿石产品，每年约采购1150万吨。

西坡项目位于澳大利亚西澳大利亚州皮尔巴拉地区，新建项目包括建造一台初级破碎机和一条与Paraburdoo加工厂相连的18公里长的输送机，将破碎后的矿石从矿山输送至帕拉布杜（Paraburdoo）现有选矿厂（帕拉布杜是力拓最早的采矿中心之一，自1972年运营至今）。

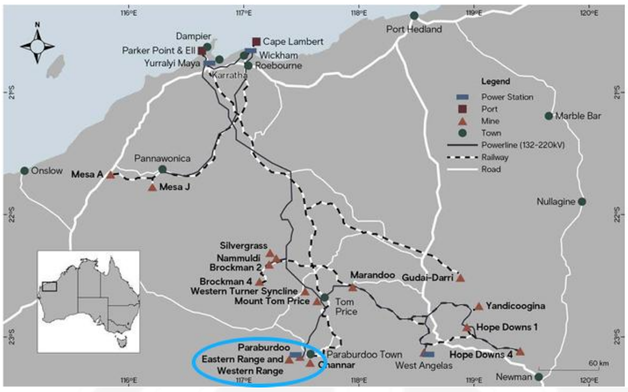
西坡项目单位现金成本预计在23-24.5美元/湿吨区间。项目依托力拓在皮尔巴拉地区的现有基础设施，单位成本具备显著竞争力，海外矿业投资介绍，西坡铁矿石的单位现金成本预计在23-24.5美元/湿吨区间，与力拓2025年皮尔巴拉地区的指导成本区间一致。

西坡项目已投产，处于产量增长过程中。项目在2025年3月首次生产，预计2025年年底前达产，届时年产能将达2500万吨，2025年剩余时间的产量继续按计划增长。

2026年力拓供给增量约250-500万吨。西坡项目主要目的是矿山替代，虽然已经投产，但是没有明显供应增量释放，大概是弥补了矿山减量的缺口，力拓披露2026年皮尔巴拉矿区将枯竭1100

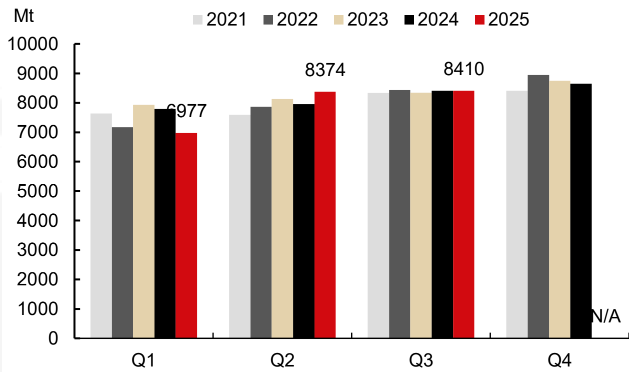
万吨产能。2026 年力拓皮尔巴拉矿区目标仍为 3.23-3.38 亿吨，与 2025 年目标一致，2025 年预计完成产量目标下限，2026 年预计发运同比增加 500 万吨，中性预期发运同比增加 250 万吨。

图表8：西坡项目位置



Source: Company's website, Mysteel, CITIC Futures

图表9：力拓三季度未见明显增量



Source: Company's website, Mysteel, CITIC Futures

2.3 Weld Range 项目

Weld Range 项目是中国宝武旗下中西矿业与澳大利亚 Fenix 公司合作开发的西澳大型赤铁矿项目，Fenix 通过支付 6000 万美元现金、后续产量提成和利润提成，获得 Weld Range 项目销售铁矿产生的 100%收益。依据 2023 年 10 月签署的《比宾-W11 矿床 1000 万吨采矿权协议》（Beebyn-W11 10Mt RTMA），Fenix 获得从 Beebyn-W11 开采 1000 万吨铁矿石的权利并无需缴纳产量提成和利润分成，2025 年 9 月签署的《威尔德岭采矿权协议》（Weld Range RTMA）则涵盖威尔德岭项目的剩余资源部分。

Weld Range 项目已赤铁矿为主，总赤铁矿资源量约 2.9 亿吨，平均铁品位 56.8%，属于中高品位赤铁矿，杂质含量低，无需复杂选矿即可直接运输销售（DSO 矿），其中 Beebyn-W11 矿产资源量估算结果为 2050 万吨，铁品位达 61.3%。项目计划整合原有产能，由 140 万吨分阶段扩产至 1000 万吨。

Weld Range 铁矿项目可共享现有基础运输设施。位于澳大利亚西澳大利亚州中西部，距离杰拉尔顿港约 500 公里，毗邻 Fenix 现有铁岭（Iron Ridge）项目，可共享物流基础设施，包括：1）Newhaul 道路物流公司（Newhaul Road Logistics），该公司拥有并运营一支最先进的公路运输车队，同时在鲁维迪尼（Ruvidini）和佩伦乔里（Perenjori）两地设有两处铁路专用线；2）Newhaul 港口物流公司（Newhaul Port Logistics），该公司在杰拉尔顿港（Geraldton Port）拥有并运营三座码头散货仓储棚，总仓储容量超过 40 万吨，年出口装载能力超过 1000 万吨。

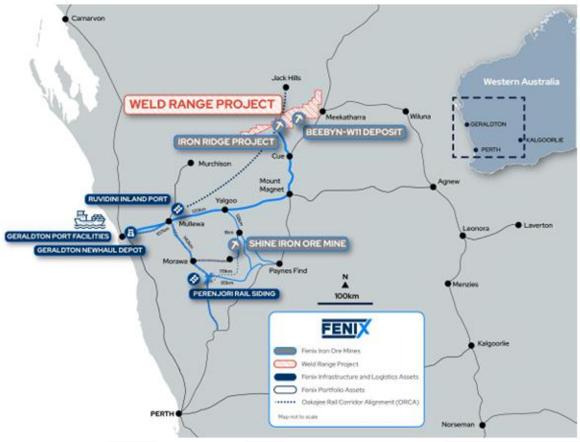
Weld Range 项目现金成本约 45-53 美元/湿吨。项目开采成本低，可共享现有基础运输设施，参考 Fenix Resources 现有生产成本，26 财年杰拉尔顿港离岸价（FOB）口径下的 C1 现金成本指引为 70-80 澳元/湿吨，约 45-53 美元/湿吨。

Weld Range 项目产能逐步兑现。2025 年 9 月，中西矿业与 Fenix 签署 30 年采矿承包协议，标志项目进入实质性开发阶段。Fenix 计划在协议签署日（Execution Date）起 12 个月内（2026 年 9 月）完成可行性研究报告，24 个月内（2027 年 9 月）启动商业生产。当前 Fenix 已启动 Weld Range 的可行性研究，该研究将整合两部分资源：一是 Fenix 目前从铁岭铁矿（Iron the Ridge Iron Ore Mine）和比宾- W11 铁矿（Beebyn-W11 Iron Ore Mine）获得的产量；二是构成 Weld Range 剩余资源的高品质赤铁矿直接装船矿（Hematite DSO）矿床。

可行性研究初期重点工作包括加快推进并拓展 Beebyn-W11 现有的采矿计划，同时在邻近的 Beebyn-W10 和 Madoonga-W14 新增采矿作业，Madoonga 矿区未来将成为 1000 万吨年产能的主力矿区。Fenix 于 2020 年 12 月启动 Iron Ridge（原自有铁矿项目，产能 140 万吨）的采矿作业，并于 2025 年 6 月启动 Beebyn-W11（Weld Range 子项目，产能 150 万吨）的采矿作业，Beebyn-W11 首批铁矿石已启动运输，2025 年 Q3Fenix 已实现 350 万吨产能，计划年内已经实现 400 万吨/年的目标产能。依据“威尔德岭采矿权协议”（Weld Range RTMA），Fenix 负有一项履约义务，即产能爬坡期结束后确保 Weld Range 的年产量维持在至少 600 万吨；同时，该公司还与宝武集团达成协议，双方将围绕“年产量 1000 万吨铁矿石的定向出口与销售”开展合作。

2026 年 Fenix 供给增量 150-250 万吨。Fenix Resources 公司的 Weld Range 铁矿项目计划产能 1000 万吨，其子项目 Beebyn-W11 铁矿石矿山 150 万吨已于 2025 年 8 月 18 日首次出货，主力矿区 Madoonga 矿区处于开发准备阶段，目前正推进项目整体可行性研究，重点整合现有基础设施与新增矿区开发方案，计划 2028 年前完成产能爬坡，实现首批 1000 万吨铁矿石出口，预估 2026 年释放增量供应 250 万吨，中性预期同比增量 150 万吨。

图表10：Weld Range 项目位置



Source: Company's website, Mysteel, CITIC Futures

图表11：Weld Range 项目资源量

Deposit	Classification	Tonnes (Mt)	Fe (%)	SiO_2 (%)	Al_2O_3 (%)	LOI (%)	P (%)	S (%)
Madoonga	Measured	63.9	57.42	6.85	2.20	7.88	0.08	0.09
	Indicated	41.9	54.98	10.14	2.13	7.95	0.08	0.14
	Inferred	13.3	54.01	12.11	2.10	7.51	0.07	0.13
	Total (Mes+Ind)	105.9	56.45	8.16	2.17	7.91	0.08	0.11
	Total (Mes+Ind+Inf)	119.1	56.18	8.60	2.17	7.86	0.08	0.11
Beebyn	Measured	58.0	58.30	6.19	2.72	6.63	0.11	0.02
	Indicated	23.6	57.21	7.56	3.07	6.53	0.10	0.02
	Inferred	7.2	54.75	9.77	4.01	6.77	0.10	0.03
	Total (Mes+Ind)	81.6	57.99	6.59	2.82	6.60	0.11	0.02
	Total (Mes+Ind+Inf)	88.8	57.72	6.85	2.92	6.62	0.11	0.02
Beebyn North	Measured	7.3	55.74	8.91	1.95	8.67	0.08	0.06
	Indicated	16.6	54.19	11.79	2.08	7.65	0.09	0.07
	Inferred	28.6	55.56	11.57	2.15	5.86	0.09	0.19
	Total (Mes+Ind)	23.9	54.66	10.91	2.04	7.96	0.09	0.07
	Total (Mes+Ind+Inf)	52.5	55.15	11.27	2.10	6.82	0.09	0.14
Madoonga West	Measured	-	-	-	-	-	-	-
	Indicated	-	-	-	-	-	-	-
	Inferred	5.3	53.81	12.68	1.91	7.90	0.12	0.03
	Total (Mes+Ind)	-	-	-	-	-	-	-
	Total (Mes+Ind+Inf)	5.3	53.81	12.68	1.91	7.90	0.12	0.03
W38/39	Measured	-	-	-	-	-	-	-
	Indicated	-	-	-	-	-	-	-
	Inferred	3.2	54.90	13.60	1.76	5.75	0.07	0.05
	Total (Mes+Ind)	-	-	-	-	-	-	-
	Total (Mes+Ind+Inf)	3.2	54.90	13.60	1.76	5.75	0.07	0.05
	Total (Mes+Ind)	232.0	57.24	7.52	2.43	7.03	0.09	0.07
	Total (Mes+Ind+Inf)	290.3	56.77	8.35	2.42	6.93	0.09	0.08

Note: 50% Fe cut-off

Source: Company's website, Mysteel, CITIC Futures

2.4 麦克菲铁矿项目

麦克菲铁矿项目是澳大利亚汉考克勘探公司旗下阿特拉斯铁矿公司开发的皮尔巴拉地区中小型铁矿项目。2025 年 7 月 1 日，汉考克勘探公司将旗下罗伊山铁矿（持股 70%）与阿特拉斯铁矿（100%）合并，成立汉考克铁矿石公司（Hancock Iron Ore）。合并后，麦克菲铁矿项目作为阿特拉斯铁矿的核心资产，纳入汉考克铁矿石公司的统一管理。

麦克菲项目投资金额约 6 亿美元，项目规划包含 5 个露天采矿坑（含部分水下矿坑），矿石资源主要为磁铁矿和赤铁矿，平均品位约 58%–60%，矿山寿命约 15 年。年产量设计为 950-970 万吨铁矿石（部分信息提及 800 万-1000 万吨），原矿无需独立选矿，需运输至罗伊山工厂加工。

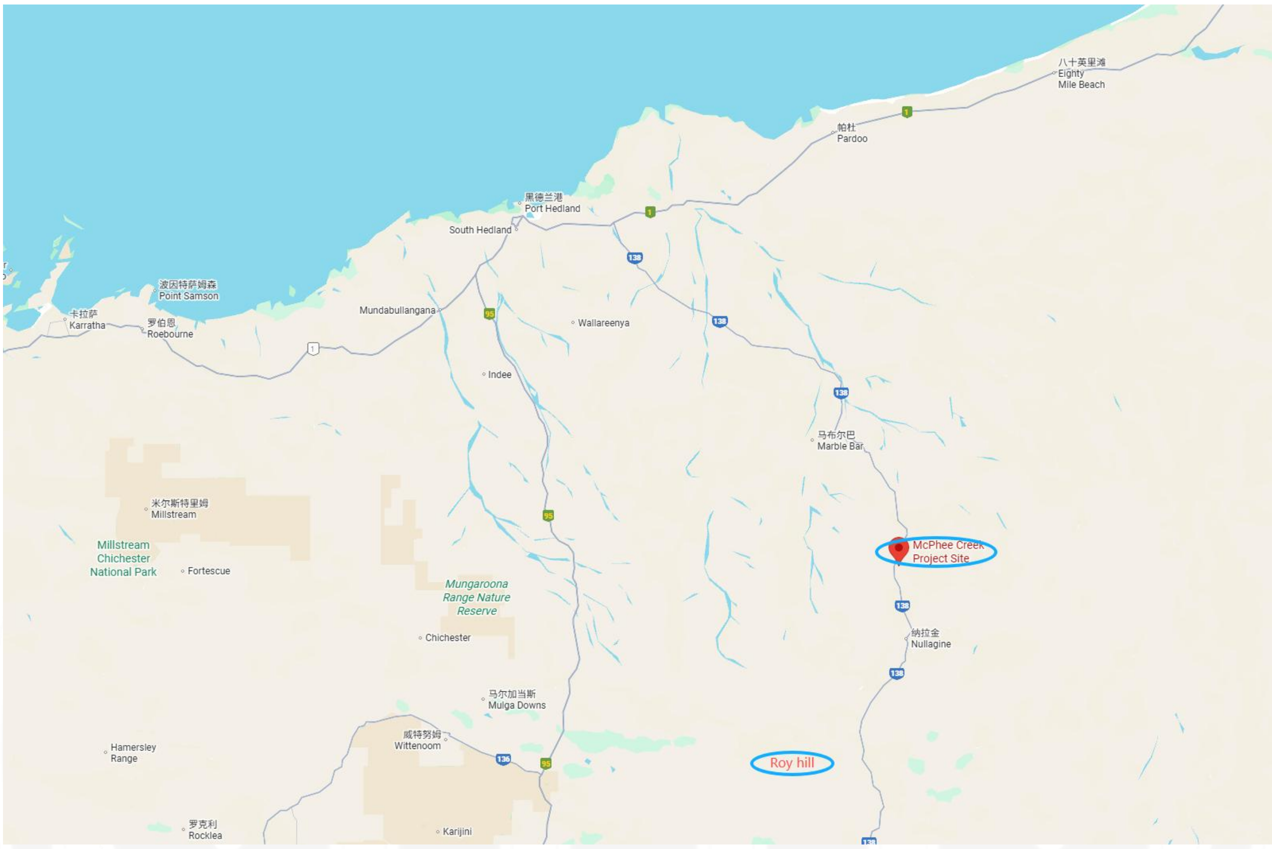
麦克菲项目可依托现有基础设施。位于澳大利亚西澳大利亚州皮尔巴拉地区东北部，距离汉考克旗下的罗伊山（Roy Hill）铁矿项目以北约 100 公里，核心特点是依托罗伊山现有基础设施实现低成本运营，靠近现有公路网络，便于接入罗伊山的物流体系。矿石开采后，在现场由矿产资源公司（Mineral Resources）旗下 CSI 服务公司运营的“年产 1000 万吨初级破碎厂”进行初步破碎，再通过公路列车运输至 100 公里外的罗伊山铁矿加工中心。依托罗伊山现有设备完成混合、脱泥等后续处理，最终通过罗伊山的 344 公里专用铁路运至黑德兰港出口，无需新建加工、铁路或港口设施。

麦克菲项目现金成本预计较低。麦克菲项目因共享罗伊山基础设施，项目省去大额基建投入，现金成本控制更具竞争力。参考罗伊山的成本水平，麦克菲项目的 C1 现金成本预计处于行业中低位（未公布具体数值，预计低于罗伊山项目）。

麦克菲项目计划近期投产但暂无落地信息。原定于 2023 年投产，因审批延误推迟，后计划 2025-26 财年出首批矿石，正式进入运营阶段，截至目前暂未看到投产信息。

2026 年麦克菲铁矿项目供应增量 250-500 万吨。若 2025 年顺利投产，有望在 2026 年释放 500 万吨供应增量，若到 2026 年中（26 财年底）投产，预计释放 250 万吨供应增量。

图表12：McPhee Creek Project 项目位置



Source: Company's website, Mysteel, CITIC Futures

2.5 Mineral Resources Limited

2.5.1 昂斯洛项目

昂斯洛项目由澳洲矿产资源公司（Mineral Resources Limited）代表 Red Hill Iron 合资企业（RHI JV）开发，MinRes 直接持股 57%，间接持股 3.3%，宝武钢铁、AMCI 和韩国浦项制铁（POSCO）分别持有 18.7%、10.7%和 10.3%的权益。项目采用"MinRes 全流程服务+合资方资源"模式：MinRes 负责项目的矿业服务与基础设施建设，包括：设计、建造并运营 150 公里专用运输公路，建设阿什伯顿港配套设施（转运码头、2 万吨级转运船），提供破碎服务等。合资方负责矿产资源所有权（MinRes、宝武、AMCI、POSCO 共同持有矿权），并通过承购协议购买项目产量。

昂斯洛铁矿项目总资源量为 7.44 亿吨，铁品位为 56.3%，项目年产能目标 3500 万吨。

昂斯洛铁矿项目位于西澳大利亚州西皮尔巴拉地区，采用 140 列巨型公路列车运输矿石，这些公

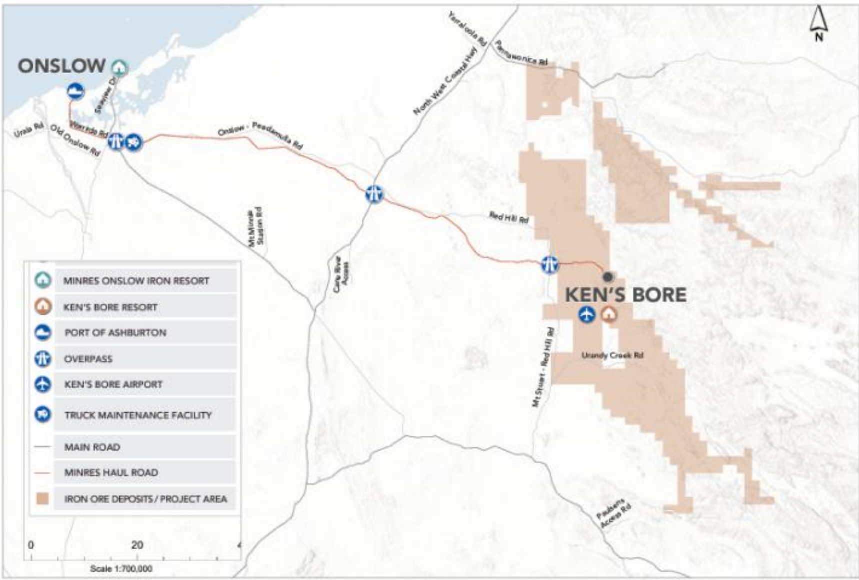
路列车长 51.6 米，载重 290 吨，从肯斯博尔矿山到阿什伯顿港的 150 公里运输路线为专用私人公路，公路列车与公共车辆没有交互作用，运输道路都设有围栏，以防止牲畜和野生动物进入道路。在港口，矿石通过封闭式转运船运输到离岸 40 公里处的等候船只。

昂斯洛项目成本约 45-54 澳元/吨，折 29-37 美元/吨。官网项目介绍资料显示，项目铁矿离岸成本（FOB）仅 45 澳元/湿吨（不含特许权使用费），2026 财年一季报公示 FOB cost 为 54 澳元/湿吨。

昂斯洛项目产能已达产，产量持续释放中。3500 万吨的产能目标原计划于 2026 年 9 月季度达成，项目于 2024 年 5 月发货，产能爬坡顺利，从当前发运数据来看，2025 年三季度已经基本达产，预计 2025 年发运 2500 万吨。

2026 年预计昂斯洛铁矿项目供应同比增加 700-1000 万吨。26 财年指引目标为 3000-3300 万吨，若 2026 年保持满产状态，预计供应同比增量 1000 万吨，中性预期供应同比增量 700 万吨。

图表13：斯洛项目位置



Source: Company's website, Mysteel, CITIC Futures

2.5.2 兰姆溪铁矿项目

兰姆溪铁矿项目是 MinRes 在西澳皮尔巴拉地区积极开发的关键增长型项目。该项目是 MinRes 皮尔巴拉枢纽的重要组成部分，是 MinRes 维持皮尔巴拉枢纽年产量（约 1000 万-1400 万吨）的关键接力棒，其核心战略目标是接力现有矿山产能，延长该枢纽的整体运营寿命。MinRes 通过其全资子公司 Process Minerals International 100%持有该项目。

兰姆溪铁矿项目产能目标约 750 万吨/年，项目资源属于层状铁矿床，是高品位的布罗克曼(Brockman)类型矿化，平均铁品位 58.3%。这种矿石通常品质较好，适合作为混合原料来提升整体产品的铁含量。它的矿石计划与铁谷（Iron Valley）的矿石进行混合，形成统一的“皮尔巴拉混合矿”产品

销售给钢厂客户。

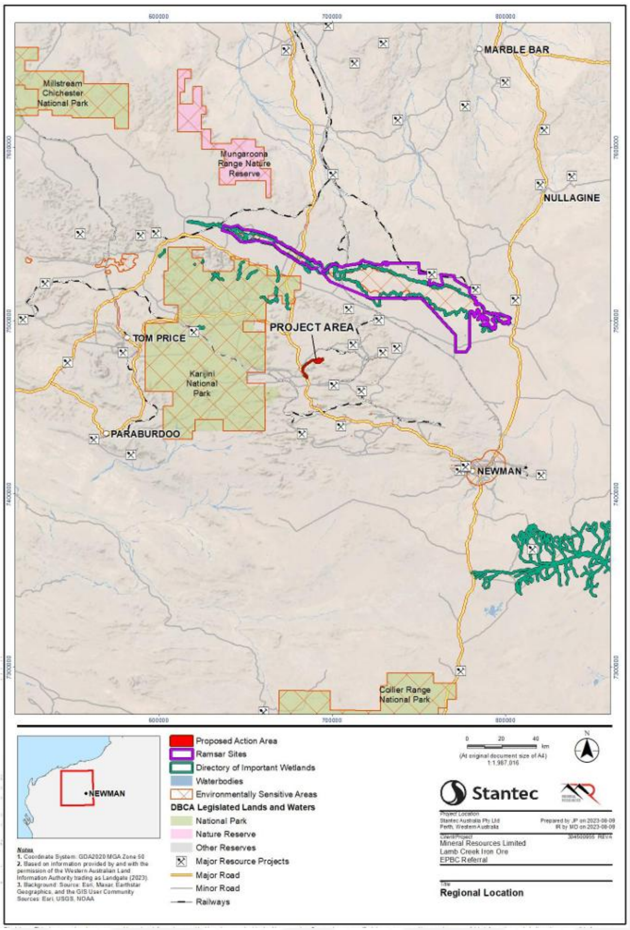
兰姆溪铁矿项目位于西澳皮尔巴拉东部，纽曼（Newman）西北约 130 公里处，距离 MinRes 现有的铁谷（Iron Valley）矿山西侧约 50 公里。矿石开采后，将在现场进行破碎和筛分。使用公路列车（Road Trains）通过约 16 公里的专用运输道路从矿区运出。接入大北高速公路，运往约 300 多公里外的黑德兰港的犹他角泊位进行出口。

兰姆溪铁矿项目现金成本预计 75-80 澳元/湿吨，折 49-54 美元/湿吨。兰姆溪项目将深度利用 MinRes 在该皮尔巴拉枢纽现有的成熟基础设施，从而大幅降低资本支出。参考 2026 财年一季报公示的皮尔巴拉枢纽铁矿石 FOB cost 为 83 澳元/湿吨，26 财年指导成本 75-80 澳元/湿吨，兰姆溪项目的铁矿石成本可能低于该水平。

兰姆溪铁矿项目将于 2026 年初开始生产。项目首次产矿时间定于 2026 年 1 月，出口时间预计在 2026 年 3 月，矿井寿命为 5 年，实际采矿时间约 3 年。

兰姆溪铁矿项目作为过渡项目，预计 2026 年不带来供应增量。兰姆溪铁矿项目的核心作用是衔接老矿区、延续皮尔巴拉枢纽的生产周期，虽然单项目来看可以贡献增量，但对 MinRes 整体而言可能不会带来产量的增加。

图表14：兰姆溪铁矿项目位置



Source: Company's website, Mysteel, CITIC Futures

图表15：兰姆溪铁矿项目投产计划表

Stage	Indicative Start	Indicative Finish	Anticipated Duration
Construction	**October 2025**	**February 2026**	**5 months**
Site access and mobilisation, construction camp	October 2025	February 2026	5 months
Site clearing, pre-strip (supporting infrastructure, mine, waste landforms, processing area)	October 2025	January 2026	4 months
Supporting infrastructure construction (e.g. internal roads, water supply and drainage control, permanent camp, WWTP, power generation, workshops, maintenance facilities, warehouse/offices)	October 2025	February 2026	5 months
Crushing and screening plant, stockpiles, RoM pad construction	November 2025	February 2026	4 months
Commissioning	**March 2026**	**May 2026**	**3 months**
Crushing and screening plant commissioning	March 2026	May 2026	3 months
Commence Mining Operation	**January 2026**	**December 2028**	**3 years**
Commence mining, haulage and export	January 2026		
Waste rock landform operation	January 2026	December 2028	3 years
Crushing and screening plant operations	March 2026	March 2029	3 years
Product export	March 2026	March 2029	3 years
Complete mining		**December 2028**	
Complete processing and export	December 2028	March 2029	4 months
Site decommissioning and closure	**March 2029**	**March 2034**	**5 years**
Post closure monitoring			Nominal 5 years

Source: Company's website, Mysteel, CITIC Futures

3. 巴西

3.1 淡水河谷

3.1.1 S11D 矿区项目

S11D 矿区是淡水河谷在巴西北部系统的核心资产，该项目为淡水河谷全资拥有和运营，项目总投资总额为 28 亿美元，是 Vale 应对全球绿色转型需求、巩固高品位铁矿石供应优势的战略基石。S11D 矿区扩建是北方系统 240 Mtpy 项目的核心，而北方系统 240 Mtpy 项目是淡水河谷新卡拉加斯计划的核心工程，由于 Serra Norte、Serra Leste 资源逐渐枯竭，为填补资源枯竭造成的空缺，北方系统 240 Mtpy 项目旨在通过整合巴西帕拉州卡拉加斯矿区的资源与技术，实现北方系统年产能 2.4 亿吨的长期目标。

S11D 矿床作为 Serra Sul 最大的矿化体，延展达 28 公里，矿层地质条件稳定，证实储量约 42.4

亿吨。该扩建项目主攻高品位矿石，铁含量保持在65%以上。结合矿区历史数据来看，实际品位大概率维持在66.7%左右的高水平。

项目位于卡纳昂-多斯卡拉雅斯市，在圣保罗市以北约1940公里处，且距离淡水河谷旗下最大的塞拉北部矿区西南方向约60公里。开采出的铁矿石经破碎后直接输送至长距离输送带，再由输送带转运至矿区内的选矿厂进行加工处理。加工后的铁矿石会先通过一条新建的101公里铁路支线运至淡水河谷的EFC铁路主干线，再借助这条铁路运往马兰哈斯州的马德里亚角海运码头。

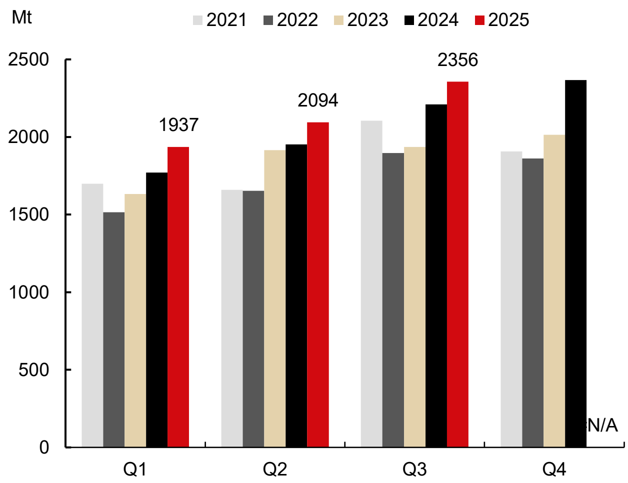
S11D矿区现金成本参考成本指导区间20.5-22美元/吨。2024年淡水河谷铁矿石整体C1现金成本为21.8美元/吨，而S11D作为公司低成本核心矿区，成本低于整体水平。2025年淡水河谷规划铁矿石C1现金成本指导区间为20.5-22美元/吨，随着S11D扩建项目推进及自动化运输、AI选矿等技术的应用，其成本有望向该区间下限靠拢。此次扩建属于在现有矿区基础上的扩产项目，无需大额新增基建投入，能进一步摊薄单位成本。

S11D矿区扩产项目预计2026年下半年投产。S11D矿区扩建（Serra Sul 120 项目，Serra Sul+20项目，S11D项目均指该项目）是北部系统S11D矿山选矿产能的扩建项目，产能目标是将S11D矿区年产能从1亿吨提升至1.2亿吨，新增2000万吨产能，预计2026年下半年投产。季报披露截至2025年7月31日，项目的财务进度已完成57%，实体工程进度达77%。

图表16：S11D 是 Serra Sul 的一部分

Source: Company's website, Mysteel, CITIC Futures

图表17：S11D 季度产量



Source: Company's website, Mysteel, CITIC Futures

3.1.2 Capanema 项目

Capanema 项目是淡水河谷在巴西东南系统实现产能复苏和可持续发展的重要项目，是淡水河谷的全资资产。该项目旨在利用一座已暂停运营的矿山，通过引入现代化的处理技术，实现低成本、环保的铁矿石生产。Capanema 项目是淡水河谷（Vale）Mariana 综合运营区的核心产能提升工程。

Capanema 项目每年能为淡水河谷新增约 1500 万吨铁矿石产能，项目开采及回收的铁矿石品质优良，含铁量达 62%，其产出的矿石主要作为烧结矿原料。

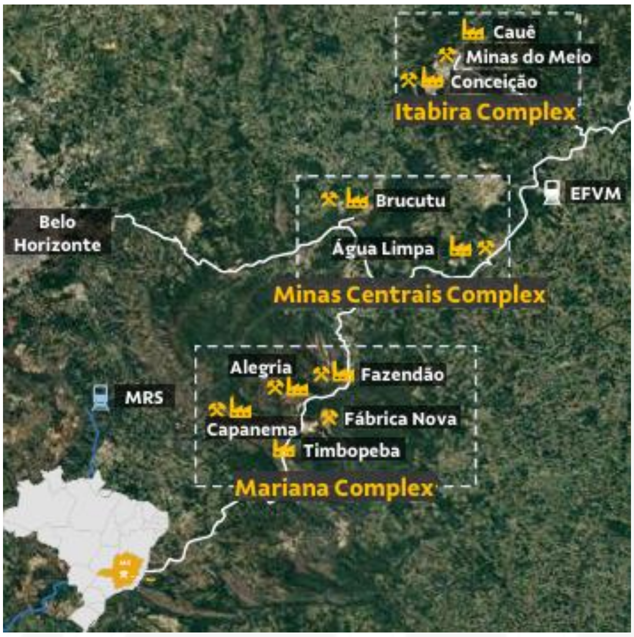
Capanema 项目地处米纳斯吉拉斯州欧鲁普雷图市，隶属于淡水河谷马里亚纳综合运营区，距离贝洛奥里藏特 80 公里，周边与圣巴巴拉、伊塔比里图等城市相邻。项目矿区内的矿石短途运输由 5 辆载重 194 吨的卡特彼勒 789D 自动驾驶矿卡负责，可实现 24 小时不间断作业。项目专门搭建了长距离带式输送机系统，高效衔接矿区开采与后续铁路运输，淡水河谷对廷博佩巴铁路货运站的产品储存堆场和装载区域进行了改造升级，以便矿石能顺畅衔接维多利亚-米纳斯铁路（EFVM）干线运输。矿石在此完成装车后，通过这条铁路专线运往圣埃斯皮里图州的图巴朗港。

Capanema 项目现金成本参考成本指导区间 20.5-22 美元/吨。Capanema 项目作为淡水河谷东南部系统的关键产能扩张项目，其铁矿石单位成本与淡水河谷 2025 年整体 C1 现金成本指导目标一致，即 20.5-22 美元/吨。Capanema 项目作为淡水河谷重点推进的绿色标杆项目，采用了诸多能降低成本的创新技术与模式，从内部比较的角度看，其生产成本大概率处于这成本区间下沿甚至更低。

Capanema 项目产能仍在爬坡。项目于 2023 年下半年启动建设，2024 年 12 月进入试运行阶段，预计 2026 年第一季度达到满负荷运行。计划 2026 年新增一条自动化选矿生产线，引入高压辊磨机技术，进一步降低能耗 15%，目标将铁品位提升至 66%。项目爬产速度较快，2025 年第二季度 Capanema 产量 60 万吨，年化 240 万吨，2025 年三季度产量达 290 万吨，年化产能已达 1160 万

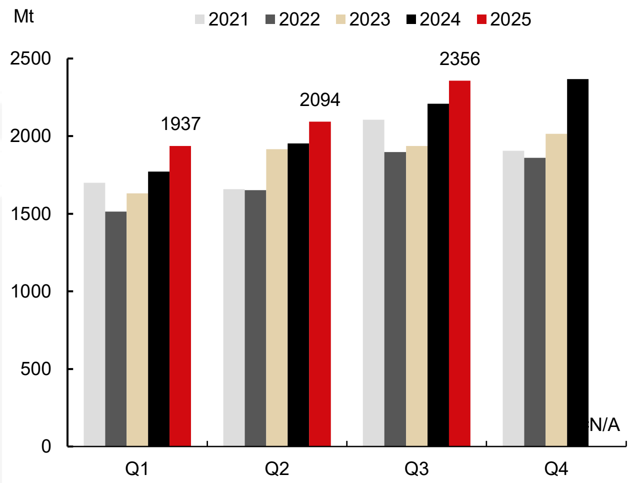
吨。

图表18：Capanema 项目和东南部系统位置



Source: Company's website, Mysteel, CITIC Futures

图表19：东南部系统季度产量



Source: Company's website, Mysteel, CITIC Futures

3.1.3 淡水河谷新项目的影响

2026 年淡水河谷供给增量预计 600-1000 万吨。此前淡水河谷预计 2026 年产量目标 3.4-3.6 亿吨，中枢较 2025 年产量目标 3.25 -3.35 亿吨增加 2000 万吨，最新公告预计（本）财年铁矿石产量为 3.35 亿公吨，并将 2026 年铁矿石产量指引下调 1000 万吨至 3.35 亿至 3.45 亿公吨，较 2025 年目标中枢增加 1000 万吨。

S11D 矿区投产时间较晚，东南部系统 Capanema 项目 1500 万吨产能仍在爬坡，符合计划预期，但产量部分被老矿区的减量对冲。新产能投放均为了对冲老矿区减量和保障实现 2026 年产量目标，实际供给同比增量可能比较有限，若完成目标下限，产量预计持平，若完成目标上限，预计产量同比增 1000 万吨，中性预期同比增加 600 万吨。

3.2 Samarco 铁矿项目

Samarco 铁矿项目是由淡水河谷与必和必拓各持股 50%的巴西大型综合铁矿项目，涵盖采矿、选矿、运输、球团加工及港口出口全产业链，曾因 2015 年尾矿坝溃坝事故暂停运营，后经技术升级与重组逐步复产，目前正推进产能全面恢复。

2024 财年底，必和必拓披露 Samarco 铁矿的资源量为 51.9 亿吨。项目开采的矿石以变质石英

赤铁矿和疏松赤铁矿为主，这类矿石的含铁品位区间在 35%-60%。该项目核心产品是铁矿石球团，这类深加工产品的品位显著提升，其球团产品铁含量约 67%。事故前项目年产能达 2600-2700 万吨，由 Alegria 和 Germano 两个矿山，三个选矿厂，三个管道，四个球团厂和一个港口组成。2015 年 11 月 5 日，Samarco 的一个尾矿坝发生溃坝事故，Samarco 的生产陷入停滞。在 2020 年 12 月 23 日 Samarco 逐步恢复运营，使用 Germano 综合矿区三座铁矿石选矿厂中的一座来选矿，并使用 Ubu 综合矿区四座球团厂中的一座来造球，年产能达到 700 万吨至 800 万吨，占萨马科总产能的 26%。Samarco 在 2025 年重启第二座选矿厂，使年产能达到约 1400 万吨至 1600 万吨，当前已基本达到阶段性复产目标。

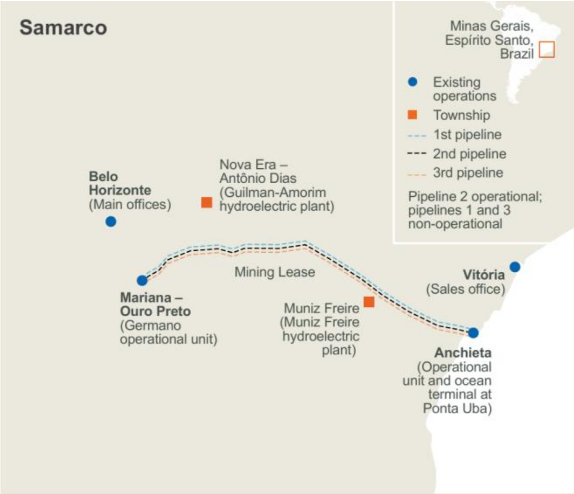
Samarco 项目拥有全球最长的铁矿精矿输送管道系统，长度约 400 公里，从米纳斯吉拉斯州的 Germano 矿区出发球团厂及港口。最初设计年输送能力 1200 万吨，经技术优化后实际运力提升至 1600 万吨，后续还通过新增增压泵站、扩建管道等方式持续提升输送效率，矿石精矿经管道输送至 Ubu 综合体加工后，直接从自有码头乌布港口装船出口全球，该海运码头可停靠载重达 21 万吨的船舶，配备一台额定装载能力约为每小时 1.2 万吨的装船机，年装船量最高可达 3300 万吨。

Samarco 项目铁矿现金成本可能显著高于必和必拓 WAIO 铁矿石的单位现金成本 18.56 美元/吨，粗略测算可能在 30-33 美元/吨。Samarco 铁矿项目运营数据常纳入必和必拓铁矿业务整体统计，必和必拓的铁矿资产主要包含西澳大利亚铁矿（WAIO）和 Samarco 铁矿两大板块，其财报中多披露铁矿业务整体或 WAIO 板块的单独成本，未将 Samarco 的生产成本单独拆分核算。2025 财年必和必拓公布 WAIO 铁矿石单位现金成本为 18.56 美元/吨，Samarco 的成本不仅包含常规采矿生产成本，还叠加了大量复产相关的额外支出，其初期成本可能明显高于该值。不完全统计，恢复初期产能已投入 2.6 亿美元，提前关闭 Germano 矿区最后的尾矿坝（Germano dam）所计划的专项资金 5.8 亿美元，Samarco 承诺投入用于恢复全部生产设施的资金达 22-28 亿美元，按满产十年折旧测算，成本增加 11.5-13.8 美元/吨，若考虑用于环境修复和灾难受害者赔偿的总支出则单位成本会更高。

Samarco 项目阶段产能恢复目标达成，产量持续释放中，未来仍有继续恢复产能的计划。2025 年三季度 Samarco100%权益产量 414 万吨，年化 1656 万吨，产能恢复至总设计产能 2600 万吨/年的 63.7%，2028 年计划实现 100%满负荷运营，达 2600-2700 万吨/年。

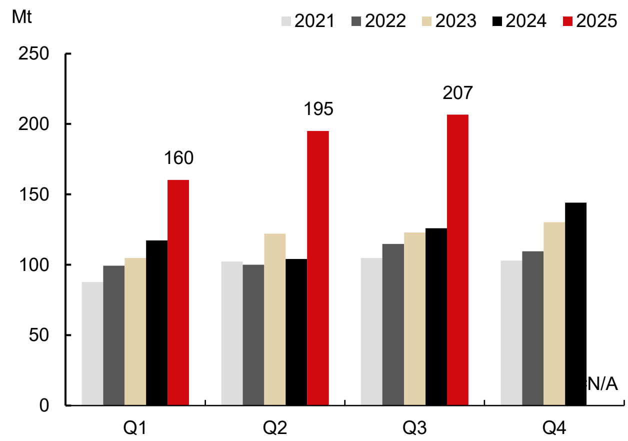
2026 年预计 Samarco 项目供应同比增量 100-150 万吨。截至 2025 年三季度 Samarco 已实现近期产能恢复目标，预计 2025 年产量 1500 万吨左右，2026 年满产情况下供应同比增量 150 万吨，中性预期同比增量 100 万吨。

图表20：Samarco 位置



Source: Company's website, Mysteel, CITIC Futures

图表21：Samarco 季度产量（BHP50%权益）



Source: Company's website, Mysteel, CITIC Futures

3.3 Pires 尾矿回收项目

CSN（巴西国家钢铁公司）的 Pires 尾矿回收项目是其在米纳斯吉拉斯州推进的核心固废循环工程，聚焦于铁矿开采遗留尾矿的资源化回收与环保处置，主要处理 Pires 矿区及周边选矿厂产生的铁矿尾矿，通过技术升级替代传统尾矿坝存储模式，实现"资源再生+环境风险管控"双重目标。

CSN 矿业公司是巴西第二大铁矿石出口商，拥有经认证的铁矿石储量超 20 亿吨，自有矿山产能约 3300-3600 万吨，加上采购矿石量，年出口铁矿石超过 4000 万吨。CSN 矿业公司计划截至 2030 年在米纳斯吉拉斯州投资 132 亿雷亚尔，用于扩大产能、提高铁矿石品位。其中 CSN 计划 2026 年一季度投产 ires 尾矿回收项目年产能 110 万吨，生产 64%品位铁矿。

CSN 持有 MRS 铁路公司股份，在巴西国内市场，该铁路承担了大部分产品的运输任务，服务于 CSN 集团自有炼钢厂，出口用铁矿石由 MRS 铁路公司从矿山铁路站点运至里约热内卢州的伊塔瓜伊港。

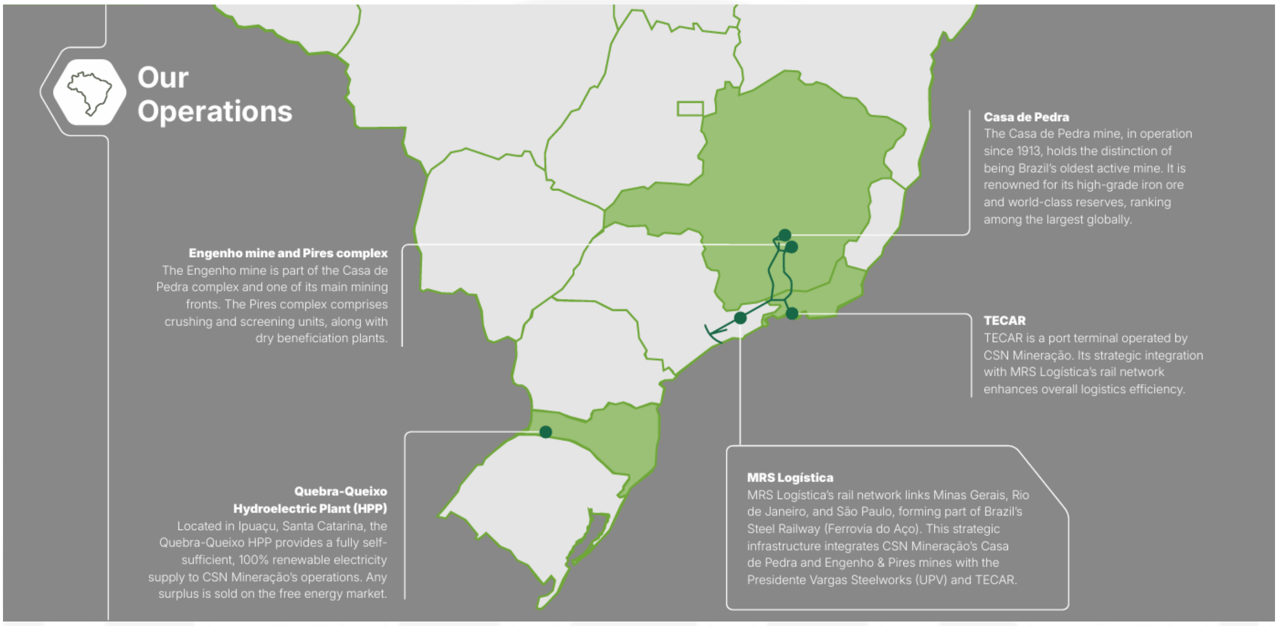
Pires 尾矿回收项目成本预计低于 20 美元/吨。参考 CSN2025 年第二季度披露的 C1 成本为每吨 20.8 美元，近两年披露 C1 成本在 20-21 美元/吨附近，尾矿回收项目主要的成本优势是消除了采矿和爆破环节的成本，使其现金成本极具竞争力，实际成本可能低于 20 美元/吨。

Pires 尾矿回收项目预计 2026 年一季度投产，其他项目在未来陆续投产。CSN 的扩产计划包括一系列项目，目标是到 2030 年将产能提升至 6000 万吨以上，核心是在孔贡哈斯市"佩德拉之家"矿综合体新建一座铁矿石选矿厂"P15 铁英岩选矿厂"，年产能 1650 万吨高品位球团矿，铁品位 67%，目前该项目正处于建设阶段，预计 2027 年四季度投产。此次投资还包括对目前储存在待退役尾矿坝中的尾矿进行回收和再利用的相关举措，CSN 计划 2026 年一季度投产 Pires 尾矿回收项目年产能 110 万吨，生产 64%品位铁矿；B4 尾矿回收项目年产能 250 万吨，计划 2027 年一季度投产，生产 66%品位铁矿；Ultrafines 粉矿回收项目年产能 100 万吨，计划 2027 年四季度投产，生产 66%品位铁矿；

Casa de Pedra 尾矿回收项目年产能 250 万吨计划 2029 年三季度投产，生产 66%品位铁矿。

Pires 尾矿回收项目预计在 2026 年带来 50-100 万吨供应增量。CSN 计划 2026 年一季度投产 ires 尾矿回收项目年产能 110 万吨，预计 2026 年带来同比增量 100 万吨左右，中性预期同比增量 50 万吨。

图表22：CSN 铁矿资源位置



Source: Company's website, Mysteel, CITIC Futures

4. 非主流国家

4.1 几内亚-西芒杜铁矿项目

西芒杜铁矿位于西非几内亚东南部，是全球储量最大、品位最高的未开发露天赤铁矿之一，也是非洲最大的矿业与基础设施一体化项目，其 1.2 亿吨的年产能将改变全球铁矿石供应格局。项目分为北部区块（1、2 号矿区）和南部区块（3、4 号矿区），由多方国际企业联合开发，中企合计持有超 50%权益，是项目核心控制方，北部区块由赢联盟（WCS）主导开发，南部区块由 Simfer 公司主导开发。

西芒杜项目总投资超 200 亿美元。设计南北区块各 6000 万吨年产能，合计 1.2 亿吨，已探明储量约 44 亿吨，总资源量预估接近 50 亿吨，平均全铁品位超 65%，且杂质含量低（如硫、磷等）。

西芒杜项目位于西非几内亚东南部内陆的凯鲁阿内（Kérouané）省。矿区位于偏远山区，距离几内亚海岸线约 600 多公里，且地形复杂。为解决矿石运输问题，项目建设了约 670 公里的铁路，跨几

内亚公司（CTG）主导基础设施（铁路、港口）的开发与运营，CTG 由赢联盟、Simfer 各持股 42.5%，几内亚政府持股 15%。初期在马瑞巴亚河口建造港口设施，矿石先经铁路运至港口，再通过驳船转运至远洋船舶出口，后续还规划建设深水港进一步提升运输效率。

西芒杜项目运营成本约 24.5 美元/吨。西芒杜项目矿体集中、埋藏浅，露天开采条件优越，开采成本低，2023 年底力拓曾披露运营成本测算约 24.5 美元/吨，但西芒杜项目前期固定投入高，基础设施分摊高，初期成本可能明显高于该水平。随着产能释放与规模效应显现，现金流成本有望逐步回落。

2026 年西芒杜项目预计贡献铁矿增量 1000-2000 万吨。12 月 3 日几内亚马瑞巴亚港满载 20 万吨高品铁矿石的远洋货轮拔锚朝着中国启航，西芒杜项目首船铁矿石成功发运。2025 年底，力拓披露 2026 年西芒杜南区（100%基准）销量目标为 500-1000 万吨，此前披露将按计划在 30 个月内逐步提升南部矿区产能，直至达到满负荷生产状态，假设北矿区进度相同，2026 年全年供应增量 1000-2000 万吨，略低于此前市场预期，当前主要是运输能力受限。

图表23：Simandou 位置

Source: Company's website, Mysteel, CITIC Futures

4.2 利比里亚-安赛乐米塔尔：宁巴铁矿二期扩建项目

安塞乐米塔尔是利比里亚铁矿绝对主导者，运营宁巴铁矿，年产量占全国产量的80%以上，是利比里亚铁矿出口的核心支柱。项目由安塞乐米塔尔集团（持有70%权益）与利比里亚政府（持有30%权益）共同合作开发，运营主体为安塞乐米塔尔利比里亚公司（AML）。需要注意的是，宁巴铁矿分为利比里亚宁巴县部分（由安赛乐米塔尔主导）和几内亚宁巴山脉部分（由艾芬豪大西洋公司主导）。

自 2005 年进入利比里亚以来，项目累计投资额已接近 30 亿美元，其中二期扩建工程单独投资 18 亿美元，是利比里亚历史上规模最大的外资项目之一。核心建设 Tokadeh 精选厂及配套设施，产能将从 500 万吨提升至 2000 万吨。安赛乐米塔尔在利比里亚宁巴县的资源量约 7.5 亿吨，主要为高品位赤铁矿，矿区铁矿基础品位在 58%-60%；二期扩建第二阶段将建造现代化选矿厂，经磁选加工后，出口的高品位浓缩矿品位可达 64%-65%。

宁巴铁矿矿位于利宁巴州，与几内亚境内的西芒杜铁矿相邻，主要通过铁路运输至布坎南港出口。耶克帕矿区至布坎南港的铁路通道是利比里亚境内唯一的铁矿石运输通道，全长 243 公里，最初由利美瑞矿业公司 LAMCO 在上世纪建成，后因内战停运。自 2005 年起，安赛乐米塔尔利比里亚公司依据矿产开发协议接管并拥有该铁路使用权，其对铁路进行了修复和升级，用于运输宁巴铁矿的铁矿石。

宁巴铁矿参考现金流成本约 30 美元/吨。宁巴铁矿为露天开采，矿石埋藏浅，开采难度低。2014 年美国地质调查局（USGS）报告中记录了安赛乐米塔尔在利比里亚生产铁矿石的成本低于 30 美元/吨，该阶段无需复杂选矿环节，生产流程简单，成本控制在行业低位水平。二期新建成的选矿厂相关设备运行、能耗及加工环节等额外增加了支出，短期可能导致吨矿运营成本显著上升，但随着产能的大量释放，未来有望逐步摊薄单位运营成本。

宁巴铁矿二期扩建项目已投产，产量逐渐释放中。安塞乐米塔尔利比里亚公司的二期扩建项目铁矿石精选厂于 2025 年 6 月正式落成投产，该铁矿精选厂是安塞乐米塔尔 18 亿美元二期扩建项目的核心部分，产能跃升由 500 万吨至 2000 万吨/年；远期规划（2026 年后）探索产能进一步提升至 3000 万吨/年的可行性，持续优化产业链与基础设施配套，同时探索生产直接还原铁（DRI）级精矿，适配全球钢铁行业低碳转型趋势，增强产品在高端市场的竞争力。

扩建项目预计给利比里亚铁矿带来同比增量 500-900 万吨。2025 年 6 月新增 1500 万吨产能，产量逐渐释放，2026 年利比里亚发运预计同比增 900 万吨，中性预期贡献增量 500 万吨。

图表24：Simandou 位置

Source: Company's website, Mysteel, CITIC Futures

4.3 南非-帕拉博拉：二期深部开发项目

Palabora 矿业有限公司（Palabora Mining Company，简称 PMC）运营的铁矿石项目，是南非唯一由中国企业（河钢资源）控股的大型铜铁共生矿项目，目前该项目是河钢资源的核心资产，也是全球罕见的多金属伴生矿开发典范。项目每年可为中国市场稳定供应约 800 万吨高品位铁矿，凭借独特的资源禀赋、低成本运营优势和稳定的中国市场导向，成为南非第三大铁矿石生产商，在全球非主流铁矿供应中占据特殊地位。

项目为全球罕见的铜-铁-磷-蛭石多金属共生矿，铁矿石资源主要分为两类：①尾矿堆存资源：截至 2025 年中磁铁矿堆存量约 1.2 亿吨，平均品位为 58%。在销售前经磁选处理加工后，含铁量可以进一步提升至最高 65%，品质适配高炉炼钢需求，仅存量资源即可支撑 10 年以上稳定生产。②深部原生资源：已探明磁铁矿总储量超 10 亿吨，与铜矿伴生（铜平均品位 0.68%-0.8%），深部矿区主井深度达 1200 米，为长期产能释放提供资源保障。原露天矿及一期地下矿设计产能 1000 万吨/年，2023 年一期工程濒临闭坑，河钢资源通过尾矿再选维持稳定产出，2024 年实际产量约 797 万吨，占南非三大铁矿生产商合计产量的 13.8%。二期扩产工程计划投入 60 亿元推进深部开发二期项目，设计产能提升至 1100 万吨/年，预计 2026 年投产后，年新增磁铁矿 300 万吨，服务年限延长 15 年。产品定位与销售渠道核心产品为高品位磁铁矿精粉（TFe 64.5%-65%），杂质含量低，适配中国、欧洲等地区高端钢铁企业的高炉炼钢需求。

项目位于南非林波波省帕拉博鲁瓦镇，矿石需通过铁路或公路运往南非东部港口，紧邻 R40 国家

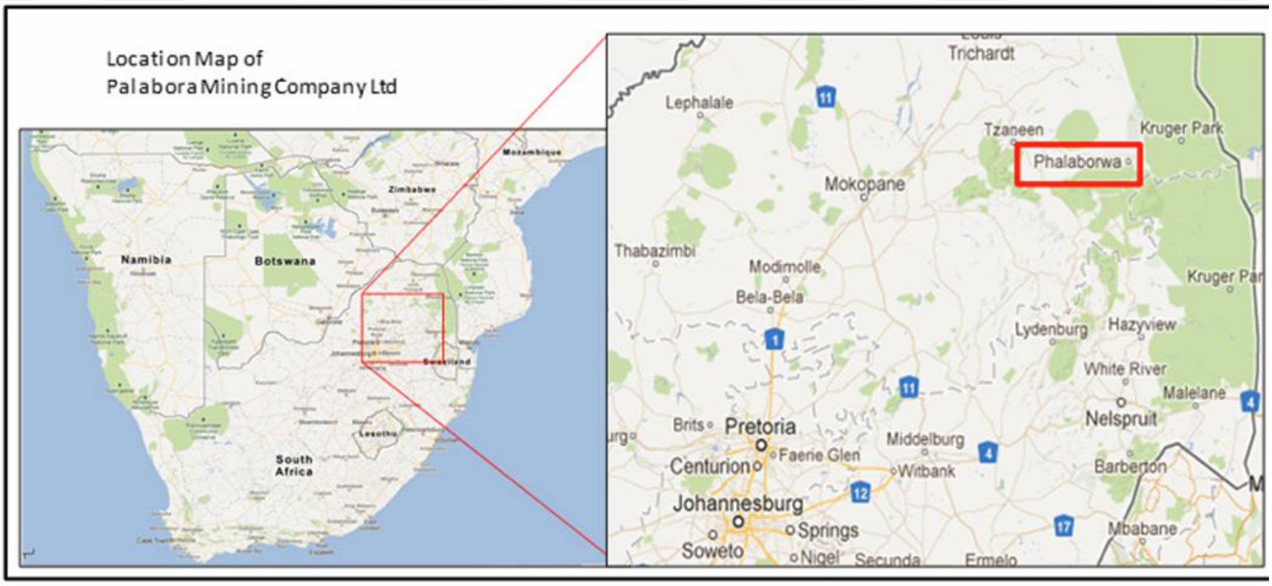
公路，可快速连接南非主要工业城市及海外港口。项目距离理查德湾港约 800 公里、莫桑比克马普托港约 400 公里。磁铁矿从 Palabora 矿区通过厂区装载铁路（或公路短倒）运至莫桑比克马普托港，理查德湾港则作为备用通道（应对马普托港拥堵或维护）。销售渠道高度集中且稳定，绝大部分产品通过四联香港公司统一承销，优先供应河钢集团内部及中国其他主流钢铁生产企业，成为中国铁矿进口的重要替代来源之一。

项目铁矿现金流成本远低于全球主流铁矿项目。公司磁铁矿是加工铜矿石过程中分离出的伴生产品，仅需要对磁铁矿进行简单的磁分离，即可将磁铁矿品位提高至最高 65%，无需额外承担开采成本，仅需精炼加工成本，相较于传统选矿的破碎、磨矿、浮选多环节流程，人力及设备维护成本大幅减少，综合采选成本远低于全球主流铁矿项目，露天矿资源枯竭后，转向地下深井开发，成本可能有所抬升。尾矿资源则通过地表堆存场直接回收，无需额外开采成本。尾矿再选模式省去勘探、采矿场地建设等前期巨额投入，单位产品固定成本通过持续产出均摊，毛利率为全球铁矿项目罕见水平，最大支出为运输成本。

项目预计 2026 年 7 月竣工及达产。达产后年新增磁铁矿 300 万吨，但其铁路运力受基础设施制约，发运效率可能影响产能释放节奏。

2026 年深部开发二期项目预计释放供给增量 100-150 万吨。2026 年 7 月竣工及达产后年新增磁铁矿 300 万吨，服务年限延长 15 年，预计 2026 年供需增量 150 万吨，中性预期贡献同比增量 100 万吨。

图表25：Palabora 位置



Source: Company's website, Mysteel, CITIC Futures

4.4 印度

4.4.1 Devadari 铁矿项目

Devadari 铁矿项目是古德雷穆克铁矿石公司（KIOCL）的核心复苏项目，是 KIOCL 自 2006 年 Kudremukh 矿关闭后首个专属自营矿山，承载着企业摆脱原料依赖、重建自主供应体系的战略使命。KIOCL 作为印度钢铁部直属的国有铁矿石出口与球团生产企业，此前主要依赖 NMDC（国营印度国家矿业开发公司）的铁矿石供应为其 Mangaluru 350 万吨/年球团厂供应原料，原料成本较高且供应稳定性不足。Devadari 项目的实施将实现原料自给，降低对第三方供应商的依赖，同时支撑 KIOCL 在 Mangaluru 的球团厂及后续选矿、造球产业的原料需求，提升公司在印度钢铁产业链中的竞争力。

Devadari 铁矿位于卡纳塔克邦 Bellary 区的 Sandur taluk，属于德干高原铁矿带的一部分。该区域铁矿石资源丰富，矿石品位约为 55%-65%，且埋藏较浅，适合露天开采。矿区总面积达 388 公顷，租约期限长达 50 年，自 2023 年 1 月 2 日与卡纳塔克邦政府矿业和地质部门签署协议并于 1 月 18 日完成注册后正式生效。项目分阶段推进投资，其中一期工程经印度公共投资委员会评估，估算总投资达 88.246 亿卢比（约合 1.06 亿美元），后续随着产能提升至 200 万吨/年，配套选矿厂的建设将新增大额投资，预计项目总投资将进一步增加。

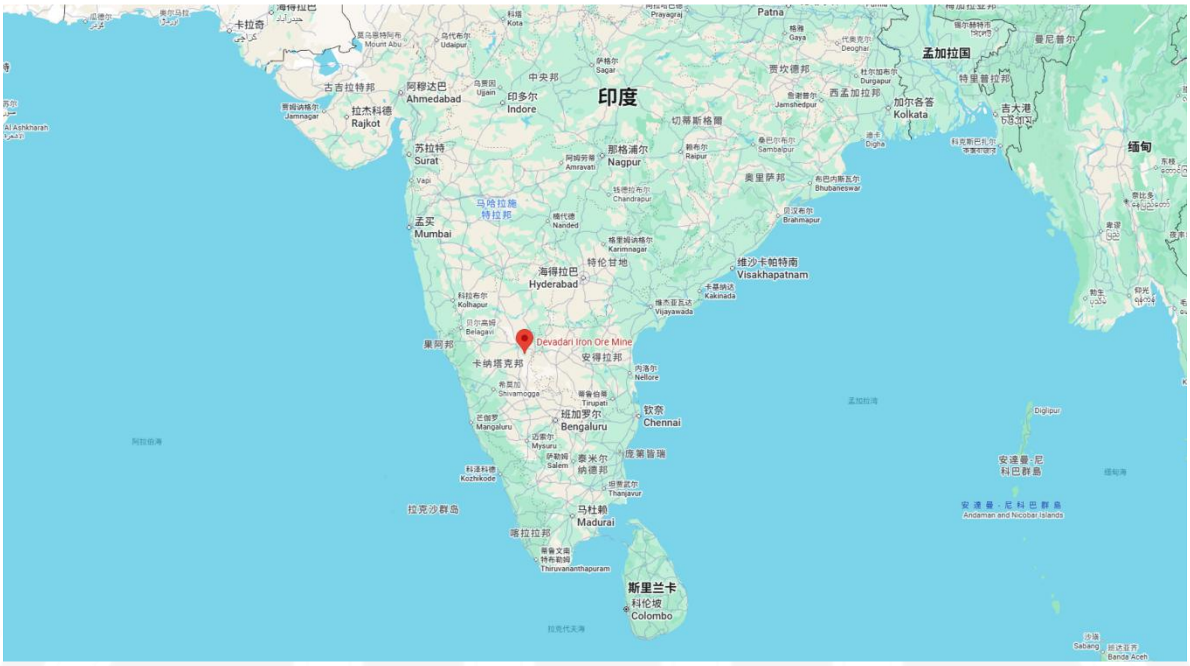
Devadari 铁矿项目精准选址于印度卡纳塔克邦贝拉里区桑杜尔乡的 Devadari 山脉区域，露天开采的原矿将通过公路运输至 Mangaluru 的球团厂（距离约 150 公里），后续将配套建设 200 万吨/年选矿厂，将原矿加工成 65%品位的铁精粉，依托卡纳塔克邦的公路网络供应国内钢厂或运往港口。

Devadari 铁矿项目主要用于国内生产。暂无项目成本相关信息，成本处于全球矿山中等水平，铁矿主要用于国内生产，出口较少。

2026 年 Devadari 铁矿项目产能计划由 50 万吨向 100 万吨迈进。项目采用"逐步爬坡、分期达产"的产能释放策略，各阶段目标明确，2024-2025 财年为试生产阶段，启动初期开采，目标年产量 30 万吨铁矿石，主要验证开采工艺、设备运行及供应链衔接效率；2025-2026 财年为产能提升阶段：产量提升至 50 万吨/年，逐步优化开采流程，扩大作业区域；2026-2027 财年为规模化生产阶段，年产量突破 100 万吨，完成主要生产设施的优化升级；2027-2028 财年达到满产运营阶段，在获得全部环境许可的前提下，实现 200 万吨/年的设计满产能力，同步建成配套选矿厂，通过选矿工艺去除脉石矿物，提升矿石品位，实现产品增值。选矿厂作为核心配套设施，其建设将基于详细勘探数据编制的项目报告（DPR）推进，计划与满产目标同步达成，建成后将具备 200 万吨/年的矿石处理能力，实现开采与加工的一体化运营。

Devadari 铁矿项目供应增量不大，2026 年预计贡献 50 万吨左右。需要注意的是，项目所在区域为生态敏感区，开采活动可能引发多重环境影响：森林与野生动物相关许可正在推进中，涉及 973 英亩区域（含 840 英亩大坝淹没区）的生态评估备案。KIOCL 承诺通过生态补偿、环保设施投入等方式降低影响，但相关措施的有效性仍需长期监测验证。

图表26：Devadari 铁矿项目位置



Source: Company's website, Mysteel, CITIC Futures

4.4.2 Surjagarh 铁矿项目

Surjagarh 铁矿是印度马哈拉施特拉邦规模最大、潜力最强的铁矿项目之一，其开采租约由印度劳埃德金属与能源有限公司（LMEL）持有。2020 年，LMEL 与瑟里维尼土方工程公司（TEPL）共同出资成立了一家名为瑟里维尼·劳埃德矿业公司（Thriveni Lloyds Mining）的合资企业，专门负责 Surjagarh 铁矿的开采运营等相关工作。其中，合作方 TEPL 持有该合资公司 60%的股权，LMEL 仅持有 40%的股权。

采矿租约至 2057 年 5 月，矿山总地质资源量达 8.57 亿吨，其中赤铁矿储量 1.56 亿吨，属于直接可售矿石（DSO），品位较高，无需复杂选矿即可满足钢铁生产需求；带状赤铁石英岩（BHQ）储量 7.01 亿吨，铁品位相对较低（30%-35%），但通过选矿技术可将品位提升至 67%，具备极高的开发利用价值。公司计划总投资 6000-6500 亿卢比（约合 72-78 亿美元），用于矿山扩产及下游产业链延伸。

Surjagarh 铁矿位于印度马哈拉施特拉邦加德奇罗利县埃塔帕利乡的 Surjagarh 村附近，地处该邦铁矿资源最富集的区域，地处印度中部，周边矿产开发基础设施逐步完善，与大多数钢铁厂距离均等，为规模化开采和直送终端提供了地理优势。项目矿石通过铁路或者管道运输直送终端，2025 年 LMEL 启用了一条 85 公里长的铁矿石浆体输送管道，该管道年输送能力达 1000 万吨，第二阶段计划将管道延伸至 195 公里。

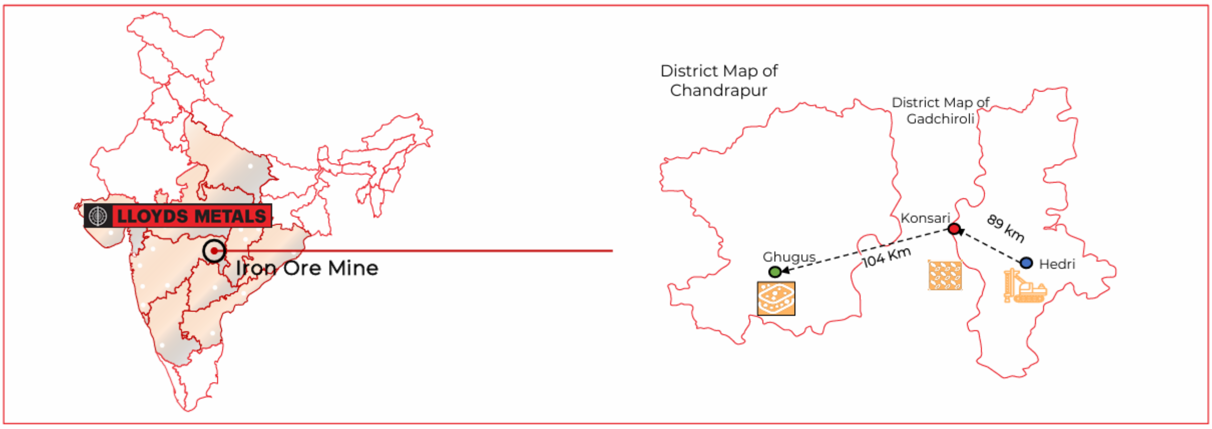
Surjagarh 铁矿项目新进来成本估算在 40-50 美元/吨附近。通过公司披露的每吨实现价减去每吨

EBITDA 得到运行现金成本（覆盖采掘、加工、运输、版税/变动税费等可变/运营成本）大概在 40-50 美元/吨附近，不含折旧/摊销、利息、一次性项目、资本性支出摊入。

近期 Surjagarh 铁矿项目产能由 1000 万吨扩产至 2600 万吨。矿山初始运营阶段产能为 300 万吨/年（原矿，ROM），2023 财年完成第一阶段扩产，产能从 300 万吨/年提升至 1000 万吨/年（原矿），配套建设了破碎和筛分工厂，年处理总量达 1176.896 万吨（含 1000 万吨原矿和 176.92 万吨尾矿），2024 财年（FY24）及 2025 财年（FY25）均维持该产能水平稳定生产。2026 财年第二阶段扩产逐步推进，2025 年 6 月，印度环境、森林与气候变化部（MOEFCC）专家评估委员会（EAC）批准产能提升至 2600 万吨/年，当前处于爬产状态；随后公司再次获得关键环境许可，明确最终扩产目标为 5500 万吨/年，该产能将整合 4500 万吨带状赤铁石英岩（BHQ）的选矿产能，随着选矿厂投产，矿山将逐步从直接销售赤铁矿转向销售选矿后的优质矿石，形成规模化、多元化的产品供应体系，2024-25 财年报提及 4-5 年实现 5500 万吨产能。扩产至 5500 万吨/年后，将成为印度产能最大的单体铁矿项目，显著提升印度铁矿自给率，减少对进口铁矿的依赖。

2026 年 Surjagarh 铁矿预计贡献增量 500-1000 万吨。LMEL 的 Surjagarh 铁矿现有 1000 万吨/年，2025 财年产量 1000 万吨，2025 年 6 月获得批准产能提升至 2600 万吨/年，计划 2026 财年开始产量将逐步爬坡至满产，生产 2000-2200 万吨，26 财年前 2 季度实际产量 740 万吨，产量释放进度可能不及预期。2026 年预计贡献增量 1000 万吨，中性预期 500 万吨。

图表27：Surjagarh 铁矿项目位置



Source: Company's website, Mysteel, CITIC Futures

4.4.3 印度果阿：拍卖重启系列项目

印度最高法院对果阿邦的全邦总产量设定了上限，目前约为 2000 万吨/年。印度果阿邦曾因大规模非法采矿引发严重环境问题与腐败丑闻，2012 年 9 月，印度最高法院应相关诉讼下达了果阿邦铁矿开采禁令。2014 年 4 月 21 日，最高法院在解除该禁令的判决中，明确设定果阿邦铁矿年开采上限

为 2000 万吨，同时要求专家委员会提交相关报告前，严格执行该产量标准。而在禁令实施前，果阿邦铁矿年出口量一度超 5000 万吨。20222024 年间，邦政府分别在 2022 年 12 月、2023 年 3 月和 2024 年 9 月完成了三轮铁矿矿区拍卖，累计拍出 12 个矿区，重启采矿产业进程中，邦政府再次强调所有运营矿区的产量总和必须控制在最高法院设定的 2000 万吨红线内，并着手制定机制，为拍卖出的矿区分配具体产能额度。

果阿邦铁矿储量预估为 10 亿吨，主要矿区集中在北果阿和南果阿的山区，包括 Bicholim、Sanguem 和 Quepem 等地区。以低品位赤铁矿为主，铁含量通常在 50%-58%之间，矿体易于开采，靠近港口，通过当地 Mormugao 港出口，具有运输和出口的地理优势。

2026 年果阿邦铁矿预计贡献增量 200-400 万吨。拍卖的 12 个矿区中，仅 Vedanta 的 Bicholim-Mulgao 矿区已经投入运营，产能 300 万吨，2024-25 财年实际产量 90 万吨，暂不确定是否已经满产。Salgaocar Shipping 的 Shirigao-Mayem 矿区已经启动运营，产能 100 万吨，暂无其产量信息。Vedanta 的 Cudnem VII 号、JSW Steel 的 Cudnem VI 号和 Kai International 的 Tivim - Pirna（VIII 号）获得了环境审批，有望在 2026 年开始运营，产能分别为 50 万吨、5 万吨、33 万吨。假设供应增量全部在 2026 年释放，预计带来同比增量 398 万吨，中性预期下，供应同比增量 200 万吨。

图表28：果阿邦位置

Source: Company's website, Mysteel, CITIC Futures

图表29：果阿邦拍卖项目情况

Auction Round	Mining Block Name	Successful Bidder	Reserves	Capacity	Progress
The First Round of Auctions (December 2022)	Bicholim-Mulgao	Vedanta Ltd.	Over 84 Mt	3 Mt	Commissioned
	Shirigao-Mayem	Salgaocar Shipping	About 24 Mt	1 Mt	Commissioned
	Monte-de-Shirigao	Rajaram Bandekar (RMB)	About 9.15 Mt	0.5 Mt	Under Permit Approval
	Kalay	Sociedade De Fomento Industries Private Limited（SFIPL）	About 16.73 Mt	1 Mt	Under Permit Approval
The Second Round of Auctions (March 2023)	Cudnem VII	Vedanta Ltd.	About 8.3 Mt	0.5 Mt	It is in the stages of signing the Mine Development and Production Agreement (MDPA), site preparation and commissioning of operations.
	Cudnem VI	JSW Steel	About 9.77 Mt	0.05 Mt	Under development; environmental permit has been obtained to date.
	Surla - Sonshi（Block IX）	JSW Steel	About 65.73 Mt	1.1 Mt	The project has entered the formal environmental approval phase.
	Adval Pale - Tivim	Fomento Resources	About 3.82 Mt	0.3-0.5 Mt	The environmental approval process is underway.
	Tivim - Pirna（Block VIII）	Kai International	About 8 Mt	0.33 Mt	Under development; environmental permit has been obtained to date.
The Third Round of Auctions (September 2024)	Onda（Block X ）	Rebidding	About 5.6 Mt	0.5 Mt	Under Permit Approval
	Curpem 和 Sulcorna（Block XI）	Rebidding	About 8.7-9 Mt	0.6-0.8 Mt	Public Hearing / ToR / EIA Review Phase
	Codli（Block XII）	JSW Steel	About 23-24 Mt	3 Mt	The environmental approval process is progressing.

Source: Company's website, Mysteel, CITIC Futures

风险提示：本文尽可能梳理了 2026 年可能带来供应增量的新增产能项目，部分公司考虑了矿山替代的影响，但不保证统计完全；投产进度在公司披露信息的基础上进行了主观推断；部分项目现金/运营成本为粗略估算，仅供参考。

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China Commodity Futures Cross Border Arbitrage Weekly Report 中国商品期货跨境套利策略周报
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人民币外汇期货风险管理应用 —— 专题报告 20251219
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20250731 China Equity Index and ETF Options Arbitrage Strategy 中国股指期权与 ETF 期权套利策略
20250728 Enhancement of Covered Call Strategy for China Equity Index Futures 中国股指期权备兑策略优化
20250626 China Gold Options Volatility and Strategies 中国黄金期权波动率及策略配置
20250311 Foreign Private Funds in China Market 外资私募进入中国市场的机遇与挑战
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20260109 Lithium Carbonate Annual Outlook Remains Positive 2026 年锂价展望：维持乐观
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20251222 The Dawn of the 61% Era for Iron Ore 铁矿石“61%时代”正在开启
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20251211 Coal-Power Long-Term Contract Interpretation 2026 年煤电长协解读
20251209 China Diesel Consumption Forecast 2026-2030 年中国柴油消费展望
20251128 China Iron Ore Futures Basis Review 中国铁矿石期货基差复盘
20251127 China Iron Ore Futures Delivery Introduction 中国铁矿石期货交割规则介绍
20251107 Solid-State Battery Introduction 固态电池基础知识
20251106 US LNG Prices May Rise on Strong Export 出口强劲或推升美国气价
20251105 Iron Ore Four Major Mines Quarterly Report Summary 铁矿石全球四大矿山季报解读
202501104 How to Understand China Gold Tax Policy Change 中国黄金税收政策调整解读
20250929 European Gasoil Crack Spread Outlook 欧洲柴油裂解价差展望
20250916 Chinese Port Inventories Pressure SC-Brent Spread 中国港口库存施压 SC-Brent 价差
20250807 Zinc Short Strategy Due To Surplus Expectation 锌过剩预期下的策略分析
20250806 Calender Spread Strategies for Lithium Carbonate Futures 碳酸锂期货月差策略推荐

【Emerging Industry】新兴行业

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20251215 New Energy Vehicle Annual Outlook 2026 年全球新能源车销量展望

20251212 Solar PV Industry Annual Outlook 2026 年光伏年度展望

20251208 Compliance Demand May Drive Carbon Prices to Rebound 履约需求或使碳价筑底反弹

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20251022 Quarterly Outlook of Energy, Power, Carbon 能源电碳行业展望

20250808 Why Anti-Involution for PV Industry 光伏反内卷原因分析

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20250902 Potential Influence of U.S. Shipping Policy Change 美国航运政策变化的可能影响

Policy Reading 政策解读

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20251210 Policy Readings on the Politburo Meeting in December 2025 年 12 月政治局会议联合点评

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20241018 Policy Readings of China SCIO Press Conference on 17th Oct. 10 月 17 日国新办发布会解学习体会

20241013 Policy Readings of China Finance Ministry Press Conference on 11th Oct. 10 月财政部发布会学习体会

【Futures 期货】

20250723 Anti-Involution Policy Reading and Influence on Futures Prices “反内卷”政策对期货价格影响

20250714 SHFE's Internationalization Reforms Interpretation 上海期货交易所国际化新规解读

20250622 China Accelerates Futures Market Opening Up 中国期货对外开放再提速

Research Framework 研究框架

【China Futures 中国期货】Macro Economy/Assets Allocation /CTA/Equity Index/ Government Bond/Exchange Rate/ HK Equity index / Hang Seng Biotech Index/Crude Oil/Gold/Copper/Iron Ore/Steels/Lithium/Silicon Metal /LPG/PTA/LLDPE/Methanol/Urea/Shipping Freight /Live Hog /Soybean Meal /Soybean Oil/Palm Oil /Rapeseed Meal/Rapeseed Oil/Corn/Natural Rubber/Zinc/Silver

宏观经济/资产配置/CTA/股指/香港股指/国债/汇率/恒生生物科技指数/原油/黄金/铜/铁矿石/钢材/锂/工业硅/LPG/PTA/塑料/甲醇/尿素/航运/生猪/豆粕/豆油/棕榈油/菜粕/菜油/玉米/天然橡胶/锌/白银

【Global Futures 海外期货】U.S. Economy / U.S. Treasury Bond/ U.S. Corn / US Dollar Index / BDI and FFAs

美国经济/美国国债/美国玉米/美元指数/航运指数

【Cross-Border Arbitrage 跨境套利】Crude Oil/Iron Ore/Soybean/Copper/Rubber/Plam Oil 原油/铁矿石/大豆/铜/橡胶/棕榈油 Cross-Border Arbitrage Statistical Regression Strategy Model 跨境套利统计回归策略模型

【Risk Management 风险管理】Iron Ore/Plastic/Steel/Freight/Fuel Oil 铁矿石/塑料/钢材/燃油/航运

【长期复盘】全球大宗商品百年回顾/全球经济展望二十年大复盘/全球炼厂利润二十年回顾 /全球炼油产能中长期展望/世界石油海运要道/国际原油价格二十年复盘/美国天然气价格二十年复盘

【国别手册】亚太：中国/印度/日本/韩国/澳大利亚/新加坡马来西亚/印度尼西亚；中东：沙特/伊朗/伊拉克/科威特/阿联酋/卡塔尔；美洲：巴西/委内瑞拉/圭亚那/加拿大；欧洲：俄罗斯/挪威/英国/法国；非洲：利比亚/尼日利亚/阿尔及利亚/安哥拉/刚果/南非/苏丹/南苏丹/埃及
Trading Manual 交易手册
Introduction on China Futures Market 中国期货市场概况 Introduction on China Futures Exchanges 中国期货交易所介绍 China Futures and Derivatives Law 中国期货法律法规 China Futures/Options Contracts Manual 中国期货合约手册 China QFII Policy Interpretation 合格境外投资者制度详解 China Futures Overseas Brokerage Policy Interpretation 中国期货境外经纪机构制度详解 Global Futures/Options Contracts Manual 海外期货/期权合约手册 **【海外期货概况】**股指/国债/汇率/能源/金属/农业 **【境外交易所概况】**中国香港/东南亚/欧洲/北美

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